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✉ For correspondence:

yosephb@upenn.edu

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Generative modeling for RNA splicing prediction and design

Di Wu¹, Natalie Maus¹, Anupama Jha², Kevin Yang³, Benjamin D Wales-McGrath³, San Jewell³, Anna Tangiyan⁵, Peter Choi^{4,5}, Jacob R Gardner¹, Yoseph Barash^{1,3} ✉

¹Department of Computer and Information Science, School of Engineering, University of Pennsylvania, Philadelphia, United States • ²Department of Genome Sciences, University of Washington, Seattle, United States • ³Department of Genetics, Perelman School of Medicine, University of Pennsylvania, Philadelphia, United States • ⁴Department of Pathology & Laboratory Medicine, Perelman School of Medicine, University of Pennsylvania, Philadelphia, United States • ⁵Division of Cancer Pathobiology, The Children's Hospital of Philadelphia, Philadelphia, United States

eLife Assessment

TrASPr is an **important** contribution that leverages transformer models focused on regulatory regions to enhance predictions of tissue-specific splicing events. The revisions strengthen the manuscript by clarifying methodology and expanding analyses across exon and intron sizes, and the evidence supporting TrASPr's predictive performance is **compelling**. This work will be of interest to researchers in computational genomics and RNA biology, offering an improved model for splicing prediction and a promising approach to RNA sequence design.

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Abstract

Alternative splicing (AS) of pre-mRNA plays a crucial role in tissue-specific gene regulation, with disease implications due to splicing defects. Predicting and manipulating AS can therefore uncover new regulatory mechanisms and aid in therapeutics design. We introduce TrASPr+BOS, a generative AI model with Bayesian Optimization for predicting and designing RNA for tissue-specific splicing outcomes. TrASPr is a multi-transformer model that can handle different types of AS events and generalize to unseen cellular conditions. It then serves as an oracle, generating labeled data to train a Bayesian Optimization for Splicing (BOS) algorithm to design RNA for condition-specific splicing outcomes. We show TrASPr+BOS outperforms existing methods, enhancing tissue-specific AUPRC by up to 1.8 fold and capturing tissue-specific regulatory elements. We validate hundreds of predicted novel tissue-specific splicing variations and confirm new regulatory elements using dCas13. We envision TrASPr+BOS as a light yet accurate method researchers can probe or adopt for specific tasks.

Introduction

Alternative splicing (AS) occurs when different mature mRNA transcripts are produced from the same gene by selectively including or excluding specific pre-mRNA exonic or intronic segments (see [Figure 1a](#) [for illustrative examples](#)). Over 90% of human genes undergo AS, with conservative estimates suggesting that at least 35% of human genes switch their dominant isoform across 16 adult tissues [Pan et al. \(2008\)](#) [Wang et al. \(2008\)](#) [González-Porta et al. \(2013\)](#) [At the molecular level, AS can alter protein function, for instance, by removing a nuclear localization signal or modifying a binding domain within the encoded protein Smith and Valcárcel \(2000\)](#) [Licatalosi and Darnell \(2010\)](#) [Alternative splicing can also serve as a mechanism to control gene expression by introducing a pre-termination codon Lewis et al. \(2003\)](#) [and aberrant splicing has been implicated in numerous diseases Singh and Cooper \(2012\)](#) [Consequently, a](#)

long-term objective for the RNA community has been to develop a predictive “splicing code” model capable of determining splicing outcomes based on genomic sequence and cell or tissue type Wang and Burge (2008) [\[1\]](#). This long-term objective forms the focus of this work.

Decades of extensive research into the mechanisms of splicing have identified hundreds of RNA-binding proteins (RBPs) involved in splicing regulation. These RBPs bind to exons and nearby intronic regions, typically within a few hundred bases of proximal splice sites, to modulate splicing in a condition-specific manner Fu and Ares Jr (2014a) [\[2\]](#). However, translating this “parts list” of RBPs into a predictive splicing code Wang and Burge (2008) [\[1\]](#) remains a significant challenge. One major technological advancement that facilitated predictive splicing code developments is high-throughput RNA sequencing, or RNA-Seq. RNA-Seq enabled researchers to detect and quantify thousands of AS events across diverse cellular conditions and tissues, as illustrated in Figure 1b [\[3\]](#). This advancement provided the data necessary to train predictive splicing codes using machine learning. Specifically, sequencing reads that span RNA segments joined by splicing (splice junction reads) are analyzed by dedicated algorithms such as MISO Katz et al. (2010) [\[4\]](#), MAJIQ Vaquero-Garcia et al. (2016 [\[5\]](#), 2023 [\[6\]](#)), and rMATS Shen et al. (2014) [\[7\]](#) to quantify these AS events.

The quantification of AS events, expressed as ‘percent spliced in’ (PSI), represents the ratio of isoforms (supported by junction-spanning reads) that include or exclude a specific RNA segment. Formally, PSI for a cassette exon e in cell type c can be denoted $\Psi_{e,c} \in [0, 1]$, and changes in splicing for that exon between two cell types c, c' , are denoted as $dPSI \Delta\Psi_{e,c,c'} \in [-1, 1]$, $\Delta\Psi_{e,c,c'} = \Psi_{e,c} - \Psi_{e,c'}$. Depending on the task, these $\Psi, \Delta\Psi$ values across numerous AS events and conditions can serve as labels to train machine learning algorithms.

As high-throughput splicing measurements became widely accessible, researchers were able to define a variety of prediction tasks based on this data. For instance, the first splicing code focused on qualitative changes in inclusion levels (“up,” “down,” or “no change”) for cassette exons across tissues, identifying associated regulatory mechanisms Barash et al. (2010) [\[8\]](#). Subsequent studies shifted to related tasks, such as predicting the effects of genetic mutations on cassette exon inclusion Xiong et al. (2015) [\[9\]](#); Cheng et al. (2021) [\[10\]](#) or the creation and disruption of splice sites caused by mutations Jaganathan et al. (2019) [\[11\]](#); Zeng and Li (2022) [\[12\]](#). In this work, we address two tasks: quantitative tissue-specific splicing prediction for a given AS event and splicing sequence design. We demonstrate how these tasks are interconnected through deep generative models and illustrate their utility. We now proceed to describe each task and the related work in detail.

The tissue-specific prediction task involves predicting $\Psi_{e,c}, \Delta\Psi_{e,c,c'}$ given the tissue or cell type and the genomic sequence of an AS event (e.g., cassette exon e). Early splicing code models relied on manually curated regulatory features from the literature to predict tissue-specific splicing changes Barash et al. (2010) [\[8\]](#); Xiong et al. (2011 [\[13\]](#), 2015). With advances in high-throughput splicing quantification, these earlier models—based on boosted decision trees and Bayesian neural networks—were replaced by deep neural networks, such as AutoEncoders, long short-term memory (LSTM) networks, and convolutional neural networks (CNNs) Jha et al. (2017) [\[14\]](#); Bretschneider et al. (2018) [\[15\]](#); Cheng et al. (2021) [\[10\]](#); Zeng and Li (2022) [\[12\]](#).

Recent years saw a shift from using predefined features and relatively small models to large models that scan wide windows of genomic sequences, typically for predicting the effect of genetic variants on splicing outcomes. Such models do not assume a specific AS event type (e.g. cassette exon) but instead look for changes in splicing within a genomic window. Most notable is SpliceAI Jaganathan et al. (2019) [\[11\]](#), which uses a CNN ResNet architecture, scanning 10 kilobases (kb) size windows across the genome to predict whether the center position is a splice site. More recently, Pangolin used the SpliceAI model architecture but trained it on several tissues and four species to predict tissue-specific PSI values ($\Psi_{e,c}$) Zeng and Li (2022) [\[12\]](#). The emphasis on large models scanning genomic windows has extended to recent Large Language Model (LLM)-based approaches for various genomic tasks. Examples include DNABERT Ji et al. (2021) [\[16\]](#), which pre-trains a BERT model on human DNA, SPLICEBERT Chen et al. (2023) [\[17\]](#), which incorporates evolutionary conservation for splice-site prediction, and Enformer Avsec et al. (2021) [\[18\]](#), which

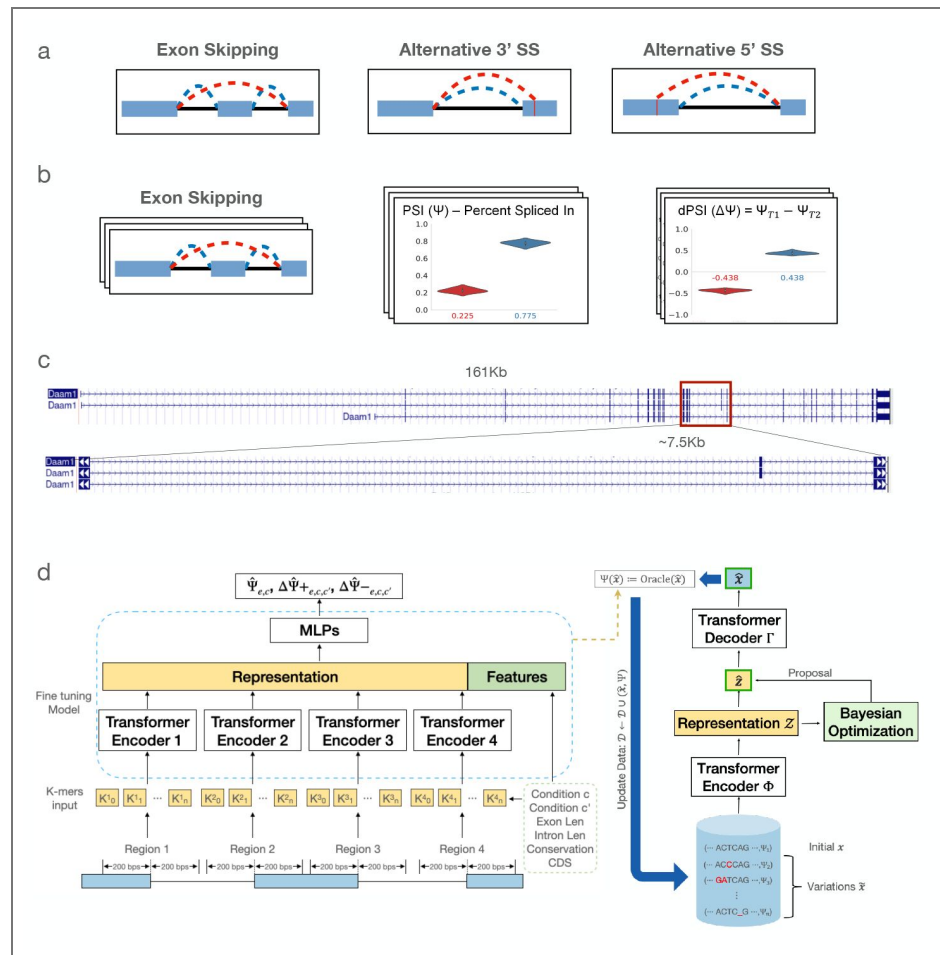


Figure 1. RNA alternative Splicing (AS) and its predictive generative modeling.

(a) cartoons of several types of AS, involving exon skipping as well as alternative 3' and 5' splice sites. (b) Quantification of AS events such as exon skipping from RNA-Seq. PSI is used to represent the relative exon inclusion level, and dPSI is used to show the inclusion change across different conditions. Quantifying thousands of such AS events across many conditions, represented by the 3D stacks, enable training ML algorithms to predict PSI and dPSI from genomic sequence. (c) A genome browser view of an illustrative exon skipping event. The genomic regions spanned by cassette exons varies from tens to hundreds of thousands of bases. (d) Schematic of TrASPr (left) and BOS (right). TrASPr's input include genomic regions proximal to splice sites around AS event e , along with other genomic features and two cellular conditions (c , c') (bottom left). It outputs a predicted inclusion level $\hat{\Psi}_{e,c}$ as well as predictions for increased inclusion and increased exclusion between the two cellular conditions $\Delta\hat{\Psi}_{e,c,c'}, \Delta\hat{\Psi}_{e,c,c'}$. Its predictions are then used to train the generative model using BOS. See main text for details.

predicts gene expression across 200-kb windows. A notable recent addition is SpliceTransformer You et al. (2024) [\[1\]](#), which we compare against and discuss in greater detail in the discussion section.

The tissue-specific splicing prediction task described above, as well as the application of LLM models to it, presents several challenges. For example, the Transformer model is a neural architecture based on self-attention that excels at modeling long-range dependencies in sequence data Vaswani et al. (2017) [\[2\]](#), which has shown strong performance in various genomic tasks. However, genomic regions containing cassette exons vary greatly in size, from a few hundred bases to several kilobases. Thus, adopting ‘out-of-the-box’ transformer-based models like DNABERT Ji et al. (2021) [\[3\]](#), which are limited to capturing sequences up to ~500 bp in length, is unsuitable for cassette exons. Even larger models like SpliceAI and Pangolin, which are highly effective at splice site detection, span no more than 10 kb. In contrast, 24% of the cassette exons analyzed in this study span a region between the flanking exons’ upstream 3’ and downstream 5’ splice sites that are larger than 10 kb. Additionally, such regions may involve multiple exons and junctions, complicating the condition-specific prediction of PSI or dPSI. Another significant challenge is sample size.

For any pair of cell types or tissues (c , c'), the number of cassette exons exhibiting significant inclusion changes is typically limited to a few hundred. Moreover, the RNA-binding proteins (RBPs) responsible for these changes are drawn from a pool of hundreds encoded in the human genome, and their specific contributions vary. Further complicating the task is the poorly defined nature of RBP binding sites, which have low information content and can either enhance or repress exon inclusion depending on their binding position Fu and Ares Jr (2014b) [\[4\]](#).

The second task we address in this work is splicing sequence design. This task can be defined as generating a genomic sequence S with a specific splicing profile across various conditions $\{c\}$. For example, the goal may be to create a cassette exon e that is highly included in the cerebellum but minimally included in other tissues. Importantly, the genomic sequence S may include intronic regions containing regulatory elements that influence the splicing of exon e . The sequence can also be based on an existing exon e that requires enhancement or correction through genetic editing. The allowed editing ‘budget’—such as the number of edits or total base changes—can be specified by the user to meet the requirements of a particular task. This flexibility is useful for applications like CRISPR editing or modifying exon inclusion using antisense oligonucleotides (ASOs) for therapeutic purposes. For instance, the model could be tasked with designing a new version of the cassette exon, restricted to no more than N edit locations (*i.e.*, start position of one or more consecutive bases) and M total base changes.

The splicing sequence design task is much newer than the tissue-specific splicing prediction task with few related work published so far. Specifically, for RNA design prior work has focused on areas such as optimizing untranslated regions (UTRs) in mRNA vaccines for enhanced expression Castillo-Hair and Seelig (2022) [\[5\]](#); Leppek et al. (2022) [\[6\]](#) and designing alternative polyadenylation sequences Bogard et al. (2019) [\[7\]](#). The algorithms employed in these studies include genetic algorithms for 5’ UTR design Sample et al. (2019) [\[8\]](#) and Deep Exploration Networks (DEN) Linder et al. (2019) [\[9\]](#). Researchers also used a greedy evolution approach where they introduced mutations one at a time to a given sequence and assessed the outcome using SpliceAI Wilkins et al. (2024) [\[10\]](#). However, none of these works defined the tissue-specific splicing design task for cassette exons as formulated here.

In the first part of this work, we address the challenges of tissue-specific splicing predictions by developing TrASPr (Transformer for Alternative Splicing Prediction), a multi-transformer-based model illustrated in Figure 1d [\[11\]](#) (left). Compared to existing models, TrASPr takes a distinct approach. While TrASPr uses the Transformer model architecture common in Large Language Models (LLM), it does not involve Billions of parameters. It also does not scan long genomic windows as common in methods such as SpliceAI and many LLM ‘foundation’ models nowadays. Instead, TrASPr uses approximately 189M parameters and leverages existing transcriptome annotations along with current knowledge of splicing regulation to focus it on specific genomic regions. Decades of research indicate that splicing regulation is primarily localized around splice

sites and influenced by condition-specific RBPs that mediate competition between splice sites. Based on this insight, we first pre-train a transformer to recognize splice sites. We then center a dedicated transformer around each of the splice sites of the cassette exon and its upstream and downstream (competing) exons (four separate transformers for four splice sites in total).

To account for features not captured by the local genomic sequence (e.g., intron and exon length, conservation), we incorporate additional genomic features. The representations learned by each transformer are combined using several joint multilayer perceptron (MLP) layers. This integrated model, TrASPr, is then trained on specific AS events (e.g., cassette exons) detected and quantified from the human transcriptome across different tissues.

By leveraging transcriptome annotations and quantifications, TrASPr can handle cassette exons spanning a wide range of window sizes—from 181 to 329,227 bases—thanks to its multi-transformer architecture. We first evaluate TrASPr using RNA splicing data from six human tissues (GTEx dataset), where it achieves state-of-the-art prediction accuracy compared to existing models. We also perform ablation studies to assess alternative architectures.

Next, we demonstrate TrASPr’s ability to detect regulatory elements, including tissue-specific ones, using diverse datasets. These include datasets involving four RBP knockdowns (KDs), a high-throughput mutagenesis assay introducing thousands of genetic variants, and a lower-throughput dataset for tissue-specific regulatory elements from a mini-gene reporter assay.

To illustrate the versatility of TrASPr, we showcase its application in two key tasks. First, we use it to predict novel tissue-specific splicing variations, which we confirm through targeted sequencing. Second, we validate TrASPr’s predictions of specific regulatory elements by targeting them with dCas13d.

TrASPr’s ability to predict tissue-specific splicing enables its use as a ‘teacher’ or ‘oracle’ for training a second deep generative model, which addresses the novel task of RNA splicing sequence design. To this end, we employ a variational autoencoder (VAE) transformer generative model, training it to structure its latent space representation using Bayesian Optimization (BO) techniques. Our Bayesian Optimization for Splicing (BOS) algorithm optimizes the VAE transformer under user defined sequence and splicing outcome constraints. We evaluate BOS-generated sequences against two baselines: random mutations and a previously proposed genetic algorithm. The results demonstrate that BOS more effectively mutates a given sequence, even with a limited number of mutations, to achieve a predefined tissue-specific splicing outcome. Additionally, we show that the mutations selected by BOS align well with experimentally validated mutations, providing further evidence of its utility. Finally, we conclude with a discussion of the potential applications of this approach, its limitations, and directions for future research.

Results

TrASPr offers improved prediction of cassette exon inclusion across diverse human tissues

To achieve accurate predictions for Ψ , $\Delta\Psi$ across diverse human tissues, we start by pre-training a 6-layer 12-head BERT model on annotated 3’ and 5’ human splice sites. We then employ four such BERT models in the architecture of TrASPr shown in Figure 1d [\[1\]](#), training it on cassette exons quantified across six human tissues using GTEx V8 (see Methods for details). To assess TrASPr prediction accuracy, we compared it to SpliceAI [Jaganathan et al. \(2019\) \[2\]](#), Pangolin [Zeng and Li \(2022\) \[3\]](#) and the recent SpliceTransformer [You et al. \(2024\) \[4\]](#). SpliceAI is considered the current state-of-the-art (SOTA) for predicting splice site strength, given a genomic window of 10 kb around it. While the model is agnostic to cassette exon definition or the cell type, it has been extensively used to approximate exon inclusion and predict mutations that disrupt splicing in patients. Pangolin adopted the SpliceAI model for tissue-specific splicing, training it on data from four species, each with four tissues [Zeng and Li \(2022\) \[3\]](#). Specifically, Pangolin was trained on quantification of 3’ and 5’ splice site ‘usage’ as defined by SpliSER [Dent et al. \(2021\) \[5\]](#) which, for cassette exons, are equivalent to MAJIQ’s LSV (local splicing variation) based PSI values. Finally,

the most recent SpliceTransformer uses a modeling approach similar to Pangolin and SpliceAI, *i.e.*, a larger genomic window (8000bp) around a 3' or 5' splice site, but replaces the ResNet CNN with a longhorn Transformer, a unique tissue specific splice site 'usage' defined by the authors, and a loss function that promotes splice sites with a tissue specific 'usage'. To match cassette exon inclusion to SpliceAI, Pangolin, and SpliceTransformer's splice sites scores we found that averaging the scores for the alternative exon's 3' and 5' splice sites performed best and applied it to the evaluations described below (see Methods).

For all models, performance was evaluated on shared tissues and test chromosomes used by Pangolin Zeng and Li (2022) [\[14\]](#) (see Data section for details). We assessed the performance of PSI prediction using two statistics: Pearson correlation (r) and the fraction of samples (denoted a) for which the model's PSI prediction was close to the observed value (< 0.2 difference, *i.e.*, within the white dashed lines in Figure 2a [\[14\]](#)). Treating SpliceAI as a baseline for SOTA non-tissue specific splice site strength prediction, we see in Figure 2a [\[14\]](#) left column that it is able to achieve $a = 0.66$, $r = 0.67$. Its tissue-specific adaptation (Pangolin) achieves $a = 0.71$, $r = 0.8$, but TrASPr outperforms both with improved PSI approximation $a = 0.78$, $r = 0.82$. Still, those statistics are dominated by extreme values, such that 33.2% are smaller than 0.15 and 56.0% are higher than 0.85. Furthermore, most cassette exons do not change between a given tissue pair (only 14.0% of the samples in the dataset, *i.e.*, a cassette exon measured across two tissues, exhibit $|\Delta\Psi| \geq 0.15$). Thus, when we repeat this analysis only for samples involving exons that exhibited a change in inclusion ($|\Delta\Psi| \geq 0.15$) between at least two tissues, performance degrades for all three models, but the differences between them become more striking (Figure 2a [\[14\]](#), right column). SpliceAI model is not tissue-specific and drops to $a = 0.47$, $r = 0.41$. As expected, the tissue-specific Pangolin model improves on SpliceAI to achieve $a = 0.55$, $r = 0.59$, but TrASPr outperforms both with $a = 0.60$, $r = 0.66$. Finally, the recently published SpliceTransformer performed similarly to SpliceAI and worse than Pangolin (see Figure 2a [\[14\]](#)), and we therefore did not consider it any further.

One potential reason for the above differences in performances is data distribution shifts. Specifically, Pangolin which exhibited the best performance compared to SpliceAI and SpliceTransformer was trained on a different algorithm's output (SpliSER) and a different dataset. To assess if data distribution was the main reason for the performance gap we first retrained Pangolin on MAJIQ's PSI and found no substantial differences in performance (pearson correlation 0.78 and 0.59 for data shown in Figure 2a [\[14\]](#) left and right column respectively). Next, instead of using the pretrained Pangolin, we tried to adopt the model and retrained it from scratch on the same data as TrASPr. Specifically, since our model only trained on human data with 6 tissues at the same time, we modified Pangolin from the original 3' and 5' splice site usage predictions to 6 PSI outputs, centering the model's genomic window on either the 3' or 5' of the cassette exon. This test resulted in low performance (3' SS: Pearson 0.21 5' SS: 0.26). This low performance may be due to various factors such as the need for much more extensive data as part of this model's training, the need to include both 3' and 5' splice site recognition in the training procedure, or lack of optimal training parameters that we were not able to identify during re-training.

Another potential contributor to the differences we observed in performance between TrASPr and the other models is the length of the region spanned by the AS cassette event. As noted above, approximately 24% of the events in our data span windows greater than the 10kb covered by Pangolin and SpliceAI, where the window is defined between the upstream exon's 5' splice site and the downstream exon's 3' splice site. Such large genomic windows where SpliceAI or Pangolin cannot "see" those flanking splice sites might lead to degraded performance of those models on those events. To test this hypothesis we divided our dataset to bins defined by the combined flanking introns length and reassessed performance for all four models. The results, shown in Figure 2 - Supp 2 [\[14\]](#), indicate TrASPr overall improved performance is not due to such long AS events, as Pangolin's pearson for events longer than 10kb was slightly better than TrASPr. This somewhat surprising result may be due to different factors, such as a different splicing mechanism for cassette exons flanked by long introns and additional information encoded by the

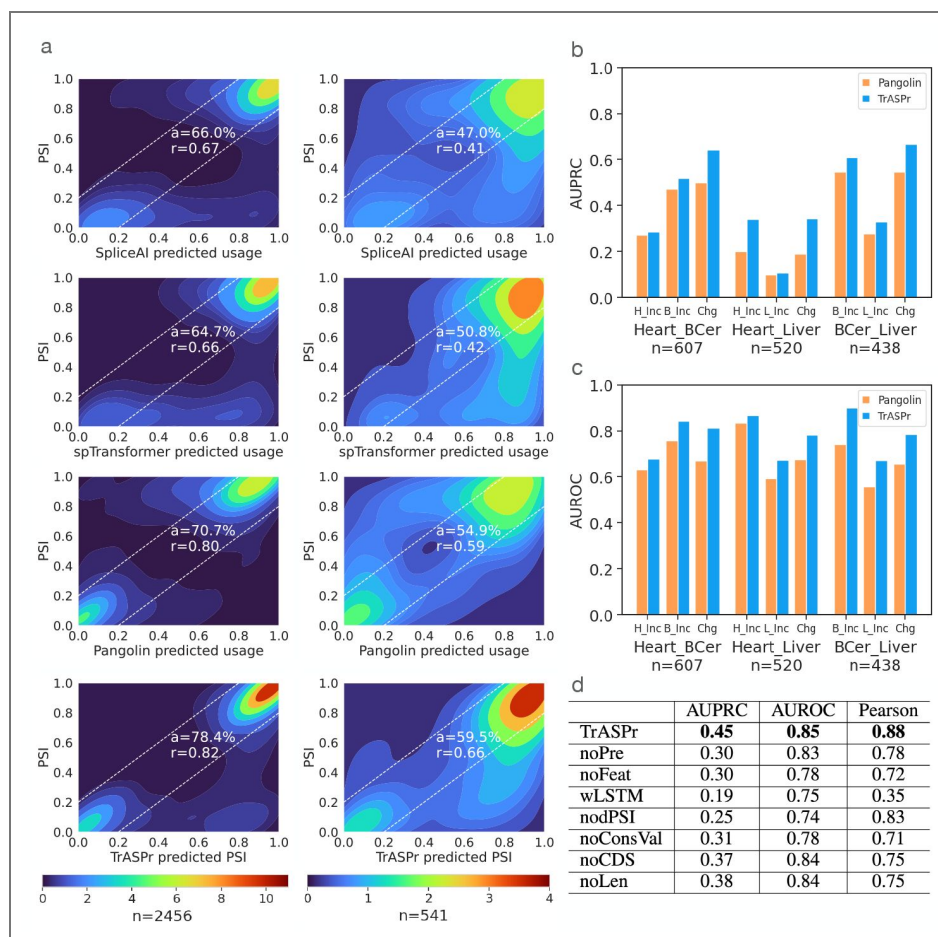


Figure 2. Comparison of PSI prediction results on GTEx dataset.

(a) Heatmaps show the distribution of prediction vs. RNA-Seq values for all samples (left) and for samples involving exons that exhibit a change ($\Delta\Psi \geq 0.15$) between at least two tissues (right) for SpliceAI (top), SpliceTransformer (2nd row), Pangolin (3rd row), and TrASPr (bottom). r is Pearson correlation, a is the proportion of predictions approximately correct (within the dashed lines). n is the number of samples (*i.e.*, a cassette event measured in two tissues) in each setting, including heart-atrial appendage, cerebellum, and liver. **(b)** AUPRC for predicting events that are differentially included ($\Delta\Psi \geq 0.15$) in one tissue (*e.g.*, heart) compared to another (*e.g.*, liver) and for predicting differentially spliced events ($|\Delta\Psi| \geq 0.15$) between the two tissues. The tissue pair is denoted at the bottom. **(c)** Same as **b** above but for AUROC. Evaluations in **a-c** include samples from tissues Pangolin was originally trained on (heart-atrial appendage, cerebellum, and liver). **(d)** Ablation study across all pairs of six GTEx tissues (the above three as well as lung, spleen, and EBV-transformed lymphocytes). AUPRC and AUROC are averaged across all tissue pairs for the change vs no-change prediction task as in **b-c**, while Pearson correlation is for PSI as in **a**. Top row (TrASPr) is the full model with pre-trained transformers. noPre - Same structure and input as TrASPr but trained from scratch. noFeat - same train/pretrain as TrASPr but without extra features. wLSTM - model with a bidirectional LSTM instead of Transformer and without the extra features. nodPSI - remove dPSIs in target function. noConsVal - same train/pretrain as TrASPr but without conservation value feature. noCDS: without coding region-related indicators and frame shifting features. noLen: without exon/intron length features.

long introns which is captured by Pangolin but not included in TrASPr input data. In contrast, when test cases are broken down by the cassette exon's length, TrASPr exhibit improved performance for longer exons (>200bp) compared to the other methods (Figure 2 - Supp 1 [1](#)).

The results above calibrate current SOTA performance for tissue-specific PSI prediction and raise a related question: Even if the exact PSI prediction is not accurate, can these models accurately predict which cassette exons are being regulated in a tissue-specific manner? Answering this question has been the focus of the original splicing code models Barash et al. (2010) [1](#); Xiong et al. (2011) [2](#); Barash et al. (2013) [3](#) and carries practical applications: Even if the magnitude of the change is not well calibrated, the models are likely able to identify regulatory features that are responsible for tissue-specific effects. In terms of assessing the models, this translates to measuring accuracy on three related *classification* rather than regression tasks of differential exon inclusion, exclusion, and change vs no-change. Specifically, given two tissues (c, c'), differentially included exons in c are exons for which $\Delta\Psi_{e,c,c'} = \Psi_{e,c} - \Psi_{e,c'} \geq 0.15$ while the negative set of exons in this classification task are exons for which $\Delta\Psi_{e,c,c'} < 0.05$. A similar definition can be made for differential inclusion in c' (or equivalently, differential exclusion in c), while the task of change vs no-change is for detecting exons that exhibit clear splicing changes, regardless of the direction of change ($|\Delta\Psi_{e,c,c'}| \geq 0.15$), compared to those that do not ($|\Delta\Psi_{e,c,c'}| < 0.05$). Figure 2b,c [4](#) summarizes the results on these three classification task in terms of AUPRC and AUROC. We note that for this task, SpliceAI is irrelevant as its predictions are not tissue-specific. The improvements offered by TrASPr for this task are particularly striking in AUPRC, which is arguably more important given the extreme label imbalance, with an average improvement of 1.28 fold and a maximum of 1.8 fold over Pangolin.

Assessing TrASPr components

The improved performance by TrASPr naturally brings the question of which components of the model contributed to this improvement. We addressed this question through a series of ablation studies where we replaced the transformer with an LSTM (wLSTM), removed the additional features (noFeat), removed pre-training (noPre), and replaced the target function (nodPSI). The results, summarized in Figure 2d [5](#), indicate each of those components significantly improved performance. Finally, we also tried to use existing DNABERT, a larger BERT model trained on the entire human genome, instead of our lighter transformer pre-trained on splice junctions. We found DNABERT to be highly unstable for this task, requiring further parameter exploration, and even then, performance degraded. These results suggest that careful pre-training can be beneficial even when a smaller model is used and that since the coding sequence is only a small fraction of the human's DNA, DNABERT may be learning dependency structures that are less relevant for the task at hand.

TrASPr generalizes to unseen cellular conditions and other AS types

Existing condition-specific splicing prediction models only generalize over unseen genomic sequences but not cellular conditions. This limitation implies that predictions were only performed for conditions for which training data already existed, limiting the usability of the splicing codes. To address this, we introduced a new component into TrASPr involving a Principle Component Analysis (PCA) that learns a latent space representation for the tissue or cellular condition (see Methods). As shown in Figure 3a [6](#), this latent space representation allows TrASPr to generalize from the six GTEx tissues to unseen conditions, including unseen GTEx tissues (Figure 3a [6](#) mid row), and ENCODE cell lines (Figure 3 - Supp 1 [7](#) mid row). Overall, using latent space representation for each tissue improves prediction accuracy compared to TrASPr lacking PCA (e.g., $a = 80.8\%$ vs $a = 79.9\%$ for GTEx tissues and $a = 78.8\%$ vs $a = 76.3\%$ for ENCODE cell lines), though naturally training on the additional GTEx and ENCODE conditions can lead to better performance (e.g., $a = 84.2\%$ for GTEx and $a = 83.4\%$ for ENCODE, Figure 3a [6](#) and Figure 3 - Supp 1 [7](#) top row). The differences become more clear again when focusing on samples involving exons exhibiting a significant splicing change ($|\Delta\Psi \geq 0.15|$) between at least two tissues (Figure 3a [6](#)

right column). Here, the six GTEx tissues model achieves a correlation of only $r = 0.39$ compared to $r = 0.53$ by the model with latent space embedding and slightly lower $r = 0.48$ for the eight tissue model. Encouraged by these results we then tested the ability of TrASPr with a latent space representation for each tissue to predict differential splicing events in those previously unseen GTEx tissues. Notably, for this task simply averaging predictions over training tissues cannot be used. The results, shown Figure 3b, indicate that, as expected, performance is improved when the tissue data is available for training ('Token' model, with train samples involving eight tissues, including the two tissues used for test samples). However using TrASPr with a PCA latent space embedding to represent tissue type can still generalize quite well to predict differential splicing in the two unseen tissues.

Next, we wanted to assess the ability of our framework to generalize to alternative splicing events which are not exon skipping. We used the same pre-trained BERT models and trained TrASPr on the same GTEx samples, but quantified junction inclusion (Ψ) for alternative 3' and alternative 5' events as shown in Figure 1a instead of cassette exons. We observed similar performance to cassette exons for TrASPr (3' $r = 0.85$, 5' $r = 0.92$, see Figure 3c). In contrast, we observed significantly worse results for Pangolin and SpliceAI on this task (Pangolin: $r = 0.36$, 0.31, SpliceAI: $r = 0.13$, 0.10 for 5' and 3' splice sites dataset respectively). It is unclear why Pangolin's and SpliceAI's performance was much worse on these tasks. Of note, to the best of our knowledge, both models were not previously evaluated on these tasks. Thus, contributing factors may be the models' focus on cryptic splice site creation (by training on splice site identification as a binary yes/no task), exon skipping being more common, or that slight changes in splice site usage between two adjacent positions are not captured well by these models. Regardless, TrASPr results demonstrate that it can generalize to both unseen cellular conditions and non-cassette exon AS events as well.

Predicting the effect of regulatory elements and mutations

Since TrASPr predicts PSI and dPSI directly from genomic sequences combined with related features, it should be able to capture the effect of both tissue-agnostic and tissue-specific regulatory elements as well as the effect of genetic mutations on those. To assess TrASPr's ability in such tasks, we first tested the effect of enhancing or degrading the core spliceosome 3' and 5' splice sites (see Data section for details). Figure 4a shows the results of this analysis for lowly included alternative exons whose splice sites were enhanced (left) and highly included exons whose splice sites were weakened (right). In both cases, the observed effect on PSI predictions is as expected, causing an increase (left) or decrease (right) in inclusion level. Notably, the magnitude of the change can vary greatly, and in general, weakening the splice sites tends to have a stronger effect. This result is to be expected as there are other elements (e.g., branch point, polypyrimidine tract) that can affect splicing, so weakening strong splice sites is likely to have a strong effect, but improving splice sites may not be sufficient to create strong exon inclusion.

Next, we reasoned that given the model's structure and the known mechanisms involved in splicing regulation, the prediction accuracy should have a strong dependency on the relative position of the learned features. To test this hypothesis, we swapped the order of the two 3' splice site transformers and the two 5' splice site transformers around the cassette exon during test time. As expected, all these pairwise swaps significantly degraded performance, with Pearson correlation on the same data used in Figure 2a dropping from 0.88 to 0.81 and 0.77 when the two 3' or 5' splice site transformers were swapped. Swapping the 3' and 5' transformers around the alternative exon abolished predictive power (Pearson correlation 0.17).

The above analysis serves as a qualitative sanity check that the model captures core splicing signals. However, a large-scale mutagenesis experiment can provide a more thorough quantitative analysis. For this, we used 6106 sequence samples where each sample may have multiple positions mutated (i.e., mutation combinations) in exon 2 of CD19 and its flanking introns and exons Cortés-López et al. (2022). The total number of mutations hitting each of the 1198 genomic positions across the 6106 sequences is shown in Figure 4b (left), while the distribution of effects ($|\Delta\Psi|$) observed across those 6106 samples is shown in Figure 4b (right). To this data we applied three

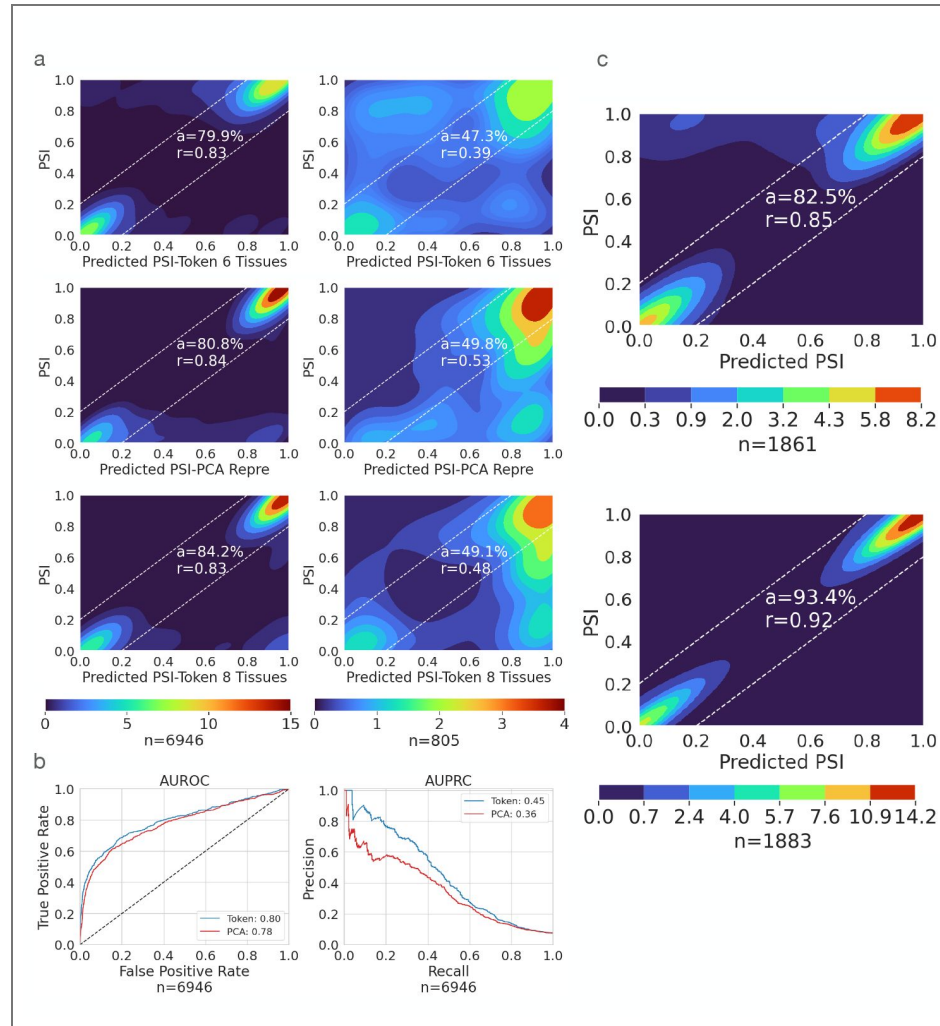


Figure 3. TrASPr prediction results in unseen conditions and alternative splice sites.

n is the number of samples in each setting. **(a)** Performance of TrASPr on two test GTEx tissues (cortex, adrenal gland). Top: The tissues were first represented as tokens, and new tissue results were predicted based on the average over conditions during training. Mid: TrASPr used the PCA learned representation to predict AS in the 2 test GTEx tissues it never trained on. Bottom: TrASPr was trained on all 8 tissues using token-based tissue representations and tested on the two test GTEx tissues. The left column includes all samples and the right one only has changing event samples. Changing events have inclusion level change larger or equal to 0.15 in at least one tissue pair. **(b)** AUROC and AUPRC plots for predicting change vs no-change events in the two GTEx test tissues used in (a), compared to the six original tissues. Blue: TrASPr with a token per tissue, trained on samples from all eight tissues. Red: TrASPr using PCA embedding to represent tissues, where samples from the two test tissues were not included in the training. **(c)** Prediction accuracy of TrASPr when applied to alternative 3' (top) and alternative 5' (bottom) splice sites.

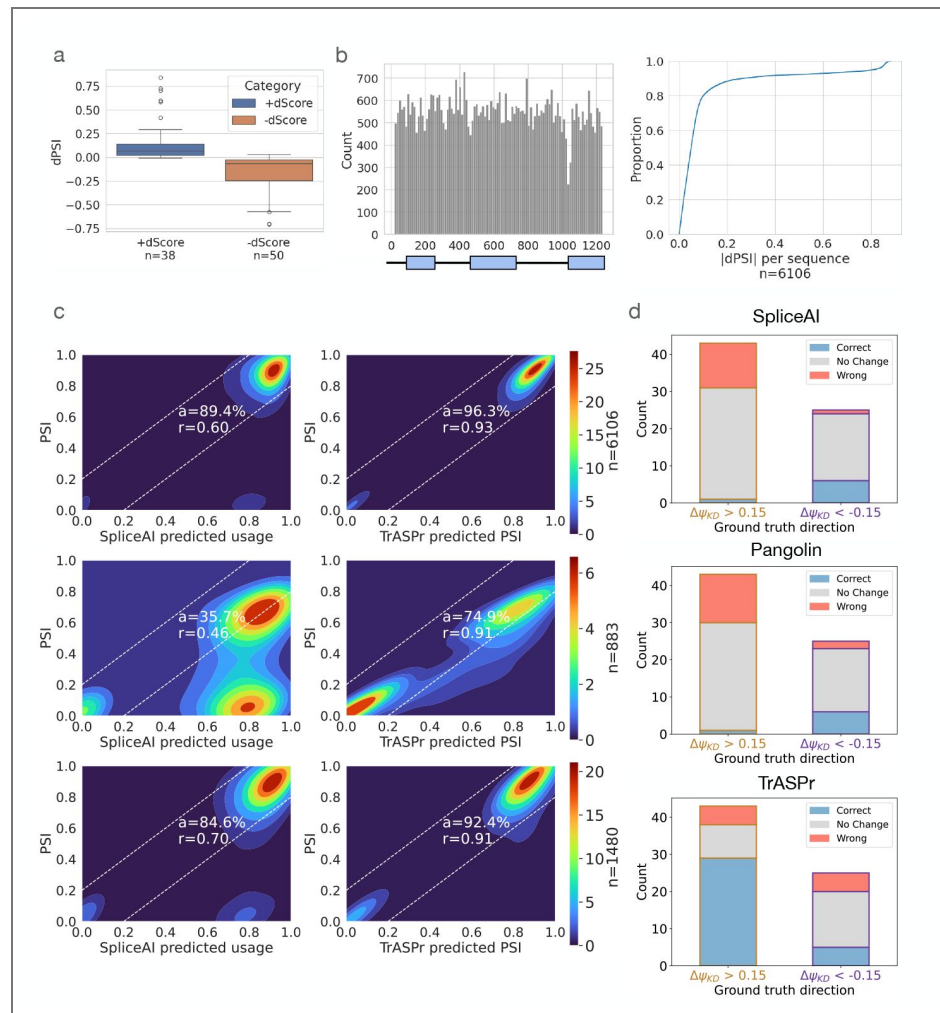


Figure 4. TrASPr prediction results on mutation effect.

(a) Whisker plot for splice site mutation effect on predicted PSI when weak splice sites are made strong (blue, left) and when strong splice sites are made weak (brown, right). (b) Distribution of mutation positions in CD19 dataset (left) and the CDF of the marginal effect per each of those sequences (right). (c) Heatmaps showing the performance of SpliceAI (left column) and TrASPr (right column) in predicting the effect of mutations shown in b, under three settings: random 5-fold cross-validation (top row), random 5-fold cross-validation for changing mutations only (middle row), and single unseen mutation filter (bottom row). n indicates the number of cases in the test set. (d) Predicting the effect (dPSI direction) of RBPs KD by mutating their corresponding sequence motifs. Blue, grey, and red correspond to correct, no change, and opposite direction prediction, respectively.

testing schemes. The first is a standard 5-fold CV where 20% of combinations of point mutations were hidden in every fold while the second test involved ‘unseen mutation’ (UM) where we hide any sample that includes mutations in specific positions for a total of 1480 test samples. As illustrated by the CDF in Figure 4b, most samples (each sample may involve multiple positions mutated) do not involve significant splicing changes. Thus, we also performed a third test using only the 883 samples where mutations cause significant changes ($|\Delta\Psi| \geq 0.15$). The results of these three testing schemes are shown in Figure 4c. As expected, TrASPr performance degrades as the testing becomes more strict, yet it significantly outperforms SpliceAI in all settings. As for Pangolin, while out of the box performance is similar to SpliceAI, retraining Pangolin on this data results in performance closer to TrASPr (see Figure 4 - Supp 1).

A caveat of the previous evaluation is that it is not focused on RBPs and tissue-specific splicing regulatory elements, with strongly affecting mutations concentrated around splice sites. To assess TrASPr predictions for RBP regulatory elements, we turned to ENCODE data. First, we assessed whether TrASPr is able to predict exon inclusion in those new conditions accurately. As shown in Figure 4 - Supp 2, TrASPr predicts Ψ within a 10% accuracy in almost 90% of the test cases, indicating excellent accuracy for the ENCODE cell lines. Next, we used RBP KD data Maurin et al. (2023); Van Nostrand et al. (2020) to compile a list of 68 cassette exons regulated by four well-studied, condition-specific RBPs (TIA1, PTBP1, QKI, RBFOX). For each of these targets, we recorded the experimentally measured effect (increased inclusion or exclusion) of the matching RBP Knockdown and compared it to the predicted effect by the model. Model prediction for the RBP KD effect was mimicked by mutating the genomic sequence corresponding to the RBP binding motif. Since most short sequence mutations are not expected to significantly alter PSI unless these ‘hit’ a core splicing signal, we set predictions to ‘no change’ when the dPSI effect was below the 95th percentile of effects observed by random mutations (see Data section for details). Overall, TrASPr performed well on most positive effect cases but predicted around half of negative effects as no change. The correlation coefficient for the dPSI effects was 0.34 and the fraction of correctly called changes was over 50%, while only 20% were called incorrectly. In comparison, SpliceAI and Pangolin performed significantly worse (Figure 4d), with both predicting correctly only for 10.3% of the cases and predicting no effect for 70.6% and 67.6% respectively. These results indicate both SpliceAI and Pangolin struggle to capture condition-specific regulatory elements.

Finally, we decided to test whether TrASPr learns positional preferences for tissue specific splice factors. For this we took 40 cassette exons exhibiting high inclusion ($\text{PSI} > 0.85$) in our GTEx tissue set, and 40 cassette exons exhibiting low inclusion ($\text{PSI} < 0.15$). We then performed *in-silico* mutagenesis analysis by inserting either a FOX binding motif (TGCATG) or a QKI motif (ACTAAC) in each possible position along the cassette exon, comparing the effect to 10 randomly chosen sequences (see Methods). The results of this analysis, shown in Figure 4 - Supp 3, recapitulate known binding maps for both splice factors with a strong preference for areas less than 150 up or downstream of the alternative exon Hall et al. (2013); Jangi et al. (2014); Van Nostrand et al. (2020). Inline with the FOX motif map Jangi et al. (2014); Gazzara et al. (2017), inclusion was most strongly promoted by FOX when the binding site was downstream of the alternative exon, and upstream when promoting exclusion. However, the strongest QKI inclusion effect occurred when the binding site was introduced up rather than downstream, contrary to what its RNA motif map would suggest Hall et al. (2013). Such effects have been reported in the literature and may be the result of context-specific effects accounted for by the model, such as additional RBPs binding in the region, but may also simply be inaccuracies of the model in terms of predicting the direction of effect in those cases.

TrASPr enables identification of new tissue specific splicing changes and regulatory elements

Given the strong performance we observed in predicting tissue specific splicing changes, we wanted to test whether TrASPr can be used to predict previously unknown splicing changes and regulatory elements. To do so, we employed LSV-Seq Yang et al. (2024), a recently-developed targeted sequencing method from our lab which allows for enrichment and quantification of

splicing events. We hypothesized that we could use TrASPr to recover tissue-specific splicing changes from previously detectable but unquantifiable low-coverage splicing events identified from analysis of GTEx RNA-Seq experiments. We first created a list of such cassette events, then assessed those for tissue-specific splicing ($\Delta\Psi > 0.1$) using TrASPr. Targeting the 787 predicted tissue-specific cassette exons predicted with LSV-Seq resulted in 558 events that had sufficient coverage (> 30 reads across at least 2/3 tissues) to be confidently quantified as changing ($\Delta\Psi > 0.1$) or non changing ($\Delta\Psi < 0.05$) compared to TrASPr predictions. Overall, compared to an expected success rate of 4.7% for random exon selection TrASPr target-selection achieved good validation rates ranging from 48.8% to 55.8% validation rate depending on the stringency of the threshold for the predicted $\Delta\Psi$ (0.1, 0.15, or 0.2, corresponding to ‘passed’, ‘stringent’, and ‘very stringent’ shown in Figure 5a [↗](#)).

Overall, the above analysis led to the identification of 169 new tissue-specific cassette exons, two of which are highlighted in Figure 5b [↗](#). The first example is of a brain cerebellum-specific cassette exon skipping event predicted by TrASPr in the ATP13A2 gene (aka PARK9). ATP13A2 is a lysosomal transmembrane cation transporter, for which loss of function mutation has been linked to early-onset of Parkinson’s Disease (PD) [Dehay et al. \(2012\) \[↗\]\(#\)](#); [Ramirez et al. \(2006\) \[↗\]\(#\)](#); [Zhang et al. \(2022\) \[↗\]\(#\)](#). Here, we detect an exon in a cytosolic loop of the protein with elevated skipping in the cerebellum. The major protein isoform with this exon skipped, which also contains variations in the C-terminal, is degraded by the proteasome after mis-localization to the endoplasmic reticulum (ER) membrane [Ugolino et al. \(2011\) \[↗\]\(#\)](#). However, the specific function of this exon in the protein remains unknown. Interestingly, many PD-associated mutations degrade ATP13A2 through a similar mechanism [Podhajski et al. \(2012\) \[↗\]\(#\)](#); [Ramirez et al. \(2006\) \[↗\]\(#\)](#); [Ugolino et al. \(2011\) \[↗\]\(#\)](#). But while ATP13A2 proteasomal degradation can have significant consequences for disease, it is unclear what role this process or its regulation by alternative splicing plays in normal tissue.

A second example of validating cerebellum-specific cassette exon skipping predicted by TrASPr involves the PTPN23 gene. PTPN23 is an essential gene with diverse molecular functions, including degradation of ubiquitinated proteins [Doyotte et al. \(2008\) \[↗\]\(#\)](#); [Ma et al. \(2015\) \[↗\]\(#\)](#), regulation of the SMN complex [Husedzinovic et al. \(2015\) \[↗\]\(#\)](#), and regulation of neuron pruning [Loncle et al. \(2015\) \[↗\]\(#\)](#). Proper regulation of PTPN23 expression is important and its dysregulation often causes disease in many tissues [Gingras et al. \(2009\) \[↗\]\(#\)](#). PTPN23 haploinsufficiency or under-expression are associated with neurodevelopmental delays [Bend et al. \(2020\) \[↗\]\(#\)](#), heart defects associated with cardiomyopathy [Xu et al. \(2024\) \[↗\]\(#\)](#), and cancer growth [Cao et al. \(1998\) \[↗\]\(#\)](#); [Manteghi et al. \(2016\) \[↗\]\(#\)](#); [Singh et al. \(2023\) \[↗\]\(#\)](#). Here, TrASPr predicts tissue specific splicing of an exon in PTPN23, whose skipping leads to a premature termination codon and nonsense mediated decay of PTPN23. This suggests a mechanism for tissue-specific regulation of PTPN23 expression, and could also be potentially used for therapeutic modulation of PTPN23 expression in disease.

Next, we turned to experimentally test specific regulatory elements. For this, we first trained TrASPr to predict differential splicing between two cell lines, HEK293T (embryonic kidney cell line) and SH-SY5Y (neuroblastoma cell line) and then selected specific positive (affecting splicing) and negative (non-affecting) regions around the cassette exon for validation (see Methods). The results for validating such regions around two cassette exons are shown in Figure 5c,d [↗](#) where g1-5 are the sgRNAs used to target dCas13 region, and Mut1,2... are the mutations selected by TrASPr based on dPSI. For all sites tested, we observed a strong agreement with the computational predictions. Taken together, these results point to the usefulness of TrASPr in predicting tissue specific splicing and regulatory elements that have not been observed or detected before. This ability raises the question if we can utilize TrASPr to then design new sequences as well. We turn to tackle this question next.

Assessing BOS sequence generation

In order to generate genomic sequence with a specific splicing profile, we use TrASPr to define a black-box objective function, and optimize it using Bayesian Optimization (BO). Specifically, using BO, we optimize over the space of possible RNA sequences to find sequences that successfully accomplish some task according to TrASPr (*e.g.*, increase exon inclusion) which serves as a teacher

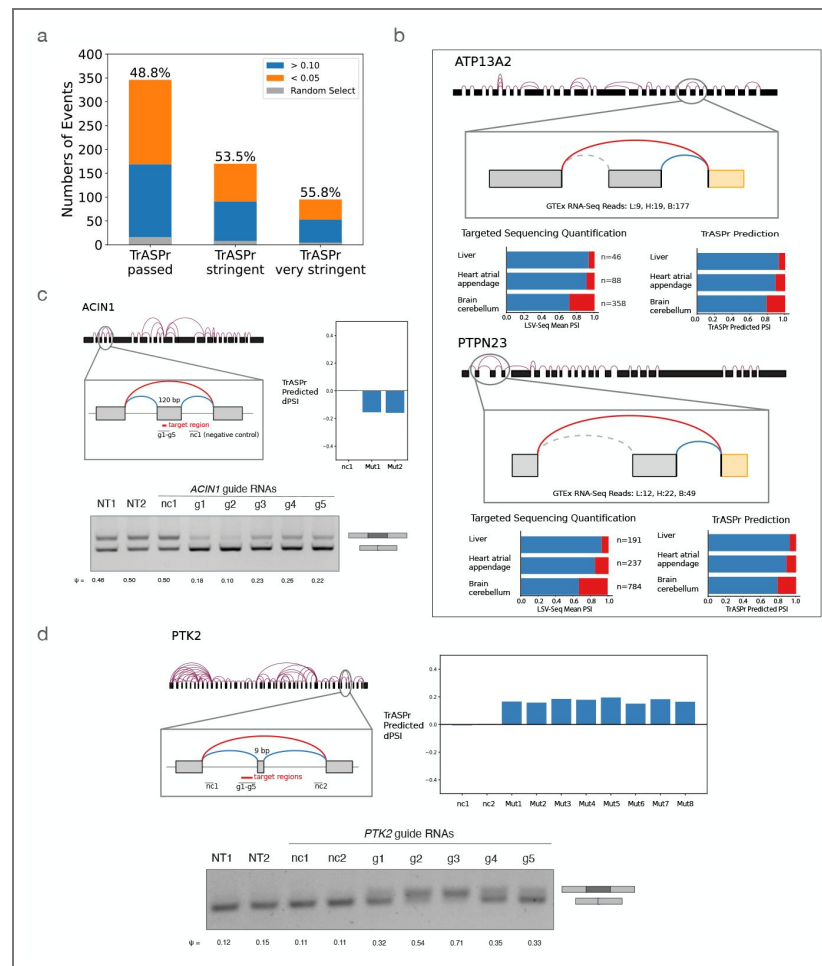


Figure 5. Experimental validations for TrASPr predictions.

(a) Bar plot for the validation rate of low coverage AS events predicted by TrASPr to exhibit tissue-specific splicing between cerebellum, liver, and heart-atrial appendage. Validation rate was between 48.8% to 55.8%, depending on the prediction stringency, discovering a total of 169 new tissue specific events. **(b)** Two examples of newly found tissue specific AS events from (a). For each case, the top graph illustrates the splicing context of the event. Two bar plots show the comparison between LSV-seq experimental results (bottom left) and TrASPr predictions (bottom right). **(c)(d)** Two AS events where specific regions were targeted by dCas13d including elements predicted by TrASPr to have significant regulatory effect and negative control regions. The bar plot (top right) shows the predicted inclusion level changes by TrASPr for 6b long windows in the tested region. Effects of dCas13d targeting were assessed by RT-PCR (bottom, NT = non-targeting, nc = negative control).

or oracle. BO is an iterative, model-based optimization procedure involving the use of a surrogate model (typically a Gaussian Process, or GP) to approximate the objective function and iteratively select the most promising candidates to evaluate. In our case, we use a GP model to iteratively select candidate RNA sequences which are then evaluated using TrASPr. Note that GPs and other standard surrogate models cannot be defined directly over the structured, combinatorial space of all possible RNA sequences. We therefore use a special kind of BO called Latent Space Bayesian Optimization (LSBO). LSBO allows BO to be applied over structured search spaces by first learning a continuous, numerical, latent space representation of the structured space. To obtain a latent space representation of the RNA sequence space, we pre-train an (unsupervised) transformer based variational auto encoder (VAE) model on a large set of RNA sequence data. We then define our GP surrogate model over the learned latent space and run LSBO. We refer to this method as Bayesian Optimization for Splicing, or BOS (see Methods for more details).

We assess the ability of BOS to generate RNA sequences for a variety of tasks and compare it to two baselines. The first baseline method randomly mutated 3, 6, 15 and 30-mers in different regions in the hope of achieving the desired effect. The second baseline is a genetic algorithm (GA) as in [Sample et al. \(2019\)](#), originally applied to design 5' UTR sequences. In all cases, we assessed how many of the generated sequences matched the desired criteria (*e.g.*, increase exon inclusion) and what was the best scoring sequence found (*e.g.*, max dPSI achieved). Note that in these evaluations we assume the values predicted by TrASPr (PSI, dPSI) for a generated sequence are correct and only assess each algorithm's ability to generate candidate sequences efficiently.

The first task we used to assess BOS was to improve inclusion levels (PSI) of weak cassette exons. Here, we started from cassette exons where either the 3' or 5' splice site was weakened and instructed BOS to improve inclusion levels. BOS was able to achieve a mean success rate of over 50% ([Figure 6a](#)) and mean increase of inclusion of 40%. In comparison both RM and GA achieved less than half of BOS success rate (~ 21%) with lower inclusion levels. Indeed, when we used the MaxEnt Yeo and Burge (2003) algorithm to score the splice sites in the generated sequences we find BOS is able to produce sequences with significantly better splice sites ([Figure 6a](#) bottom).

While the above results indicated strong performance by BOS they relied on computationally assessing performance by TrASPr and the MaxEnt algorithm. To assess BOS generated sequences with respect to experimental evidence we employed the high-throughput mutagenesis CD19 exon 2 experiment [Cortés-López et al. \(2022\)](#). Overall, this experiment included mutations to 1198 positions spanning the entire cassette exon region as shown in [Figure 4b](#). To assess site selection by BOS we computed the marginal effect observed per position when mutated and compared it against BOS frequency of selecting it. Plotting the location of mutated positions to decrease exon2 inclusion we see BOS 'locks' on the two areas proximal to the splice sites which have the largest marginal effect ([Figure 6b](#)). More generally, we find that BOS learns to favor more affecting positions in terms of their marginal dPSI effect, selecting the top 10% of affecting positions (marginal dPSI > 0.114) 31.2% compared to only 11.0% for positions at the bottom 10% (dPSI < 0.03, binomial value < 10^{-4}).

Next, to assess the ability to alter splicing events in a tissue specific manner, we supplied each algorithm with lowly included cassette exons and requested no more than 30 edits that will make those have a relative increased inclusion in cerebellum compared to other tissues of at least 30%. On this task, we considered generated sequences with high inclusion level in cerebellum (PSI > 0.5) and low inclusion in other tissues (PSI < 0.2) as successful. Notably, this is a much harder task compared to the non-tissue specific ones described before. Indeed, when assessing ten different lowly included exons as a starting point, TrASPr was only able to match the user constraints 20% of the time on average, but GA and RA exhibited a much lower success rate of approximately 4% ([Figure 6c](#) top). The differences in performance between the algorithms was even more striking when considering the best 20% of proposed sequences ([Figure 6c](#) bottom). Finally, when we assessed the contribution of intronic mutations (more than 8 bp away from the nearest splice site) introduced by BOS, we found these contributed most of the neuronal specific signal, with more pronounced effects achieved by the intronic upstream mutations ([Figure 6 - Supp 1](#)).

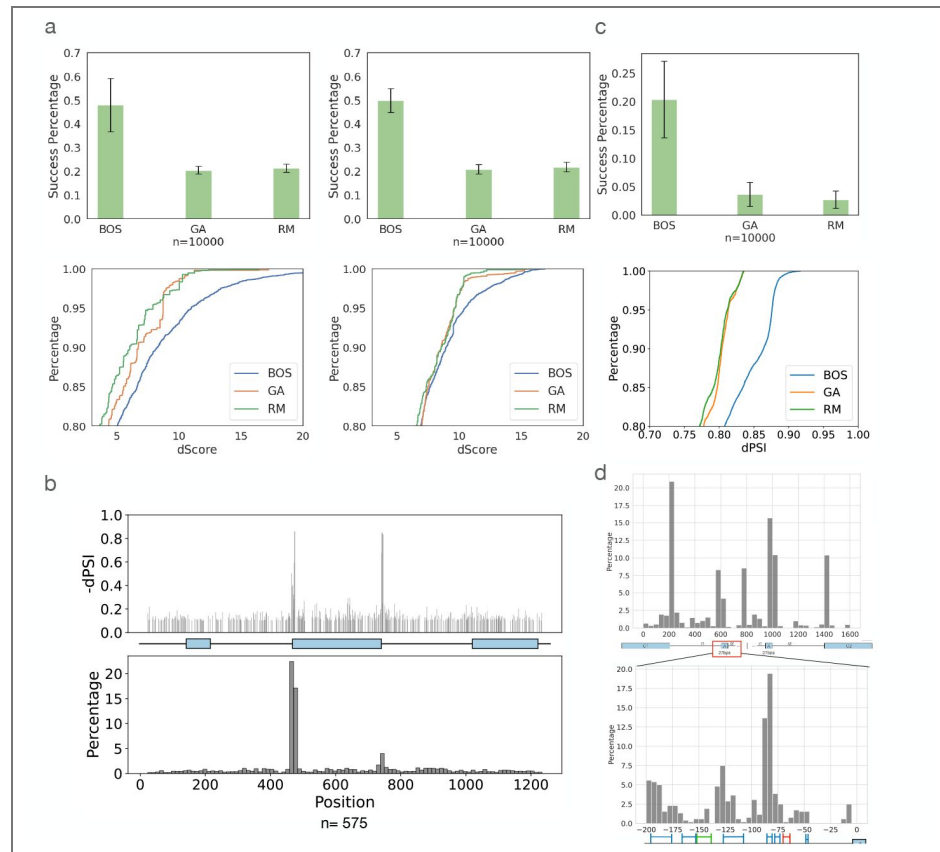


Figure 6. RNA design results by BOS.

(a) Results for the task of improving inclusion of weak cassette exons ($n=8$ exons). Top: Bar plots for success rate in achieving desired design task (increased inclusion). Error bars represent standard deviation over the set of exons tested. Bottom: CDFs over the best designed sequences (top 20%) by the MaxEnt splice site score change between the original sequence and proposed sequence. GA - Genetic Algorithm, RM - Random. **(b)** BOS generation results for CD19 mutation dataset, showing 575 generated sequences across 4846 possible positions. The positions mutated by BOS (bottom) capture regions close to the alternative exon splice sites whose mutations have strong marginal effects on inclusion levels (top). **(c)** Comparison of BOS, GA and RM on tissue-specific(cerebellum) sequence generation. Different start sequences ($n=10$) are randomly chosen from cassette exons exhibiting low inclusion levels. Every algorithm is tasked with adopting the start sequence to achieve cerebellum-specific high-inclusion ($\Psi \geq 0.5$ for Cerebellum, otherwise $\Psi \leq 0.2$) within 30 edits. Top: Success rate for this task. Bottom: The achieved improvement (dPSI) for the top 20% sequences generated by each algorithm. **(d)** BOS generation results for neuronal specific Daam1 exon 16. Bar plots indicate the distribution of 27206 hits where BOS mutated from 2000 sequences. The bottom plot is the zoom-in region of the top one. Regions that were validated experimentally by mutating them in a mini-gene systems are marked either blue (yes) or red (no) depending if TrASPr that teaches BOS is able to predict the effect of those segments. The green region indicates a region that doesn't affect the inclusion level and is predicted correctly by TrASPr.

Finally, to assess BOS generation with respect to experimentally tested tissue specific regulation we gave BOS the sequences proximal to exon 16 in the Daam1 gene and instructed it to reduce inclusion levels but without abolishing exon inclusion ($\text{PSI} > 0.1$). Exon 16 is a well studied, neuronal specific, micro-exon which has been shown to have high inclusion levels in cerebellum and in N2A cell lines ($\Psi = 0.66$). As such, previous work using a mini-gene reporter assay in N2A cells mapped several regulatory elements in the upstream intron that affect its inclusion (Figure 6d, colored regions). On this task, the best random mutation setting (30-mers) successfully generated 177 out of 4392 sequences (4.03%) that increased inclusion by more than 0.2. The GA successfully generated 210 sequences (4.7%), while BOS generated 1331 successful sequences (30.3%) that matched the constraint of $\text{dPSI} > 0.2$ (See Figure 6 - Supp 2). Furthermore, BOS also significantly outperformed the two baselines in terms of the best candidate sequence generated (0.71 maximum dPSI, compared to 0.61 and 0.53 for the GA and the random sampling algorithms, respectively). Majority of positions BOS chose to edit were, as expected, around the splice sites (Figure 6d top). However, zooming in on the upstream intron (Figure 6d bottom), we found BOS repeatedly mutated the validated enhancing regulatory elements, avoiding the negative control region (green). As expected, BOS still failed to suggest editing the red region on which TrASPr itself fails, demonstrating that the generative process is inherently limited by the capabilities of the oracle.

Overall, our analysis of *in-silico* predictions and experimental assays indicate that BOS is able to efficiently capture regulatory elements in a given sequence, including both splice site signals as well as deep intronic elements, then capitalize on those to generate sequences matching a given splicing target function.

Discussion

In this study, we offer two main contributions. First, we propose a new tissue-specific splicing code model, TrASPr. TrASPr's architecture leverages the transformer attention mechanism while utilizing multiple transformers, each focused on specific genomic regions. This design ensures the model concentrates on areas most relevant to splicing regulation without requiring excessively large models and training datasets.

We demonstrated that TrASPr significantly outperforms current state-of-the-art models in predicting PSI and dPSI across multiple datasets, even when those models used considerably larger genomic windows. Moreover, to our knowledge, this is the first demonstration of the ability to predict PSI and dPSI under previously unseen experimental conditions and to predict variations in both 3' and 5' splice sites. Using TrASPr, we generated predictions that were experimentally validated to identify tissue-specific splicing variations undetectable by previous RNA sequencing, as well as condition-specific regulatory elements.

In terms of related work, the models closest to TrASPr are Pangolin Zeng and Li (2022) and the recent SpliceTransformer You et al. (2024), both designed to predict tissue-specific splicing. Splice-Transformer employs a similar approach to SpliceAI and Pangolin, scoring genomic positions as 3' or 5' splice sites within a window spanning 4,000 nucleotides upstream and downstream. Additionally, SpliceTransformer incorporates what the authors define as splice site “usage” across different GTEx tissues. Adding this usage statistic into its target function is designed to aid in tissue-specific splicing prediction. However, in practice we found that SpliceTransformer performed similarly to SpliceAI and worse than Pangolin (see Figure 2a). A potential reason for this may lie in the Splice-Transformer's “usage” based target function. “Usage” is defined in You et al. (2024) as the fraction of GTEx tissue samples in which a splice site was identified. For instance, a value of 0.1 indicates the splice site was detected (i.e., supported by more than one read) in 10% of the tissue samples. Consequently, this addition in the SpliceTransformer's model to SpliceAI's original target function correlates poorly with splicing or differential splicing quantification. Instead, “usage” primarily reflects splice site detection capability, which is largely influenced by read depth and gene expression levels (see Figure 2 - Supp 3). We note that these results do not negate SpliceTransformer's ability to detect mutations that disrupt or create splice sites for downstream analysis, similar to SpliceAI and Pangolin.

The second key contribution of this study is the formulation of RNA sequence design with specific splicing characteristics as a Bayesian optimization problem. We developed the BOS algorithm, which uses TrASPr as an oracle to tackle this design challenge by introducing biologically plausible mutations. Across various tasks, we showed that BOS effectively generates RNA sequences with the desired splicing changes, incorporating mutations that selectively create or disrupt core splicing signals and intronic regulatory elements as required. Here we compared BOS to a previously used genetic algorithm and randomly introduced mutations as a baseline. However, more recent publications proposed additional methods to generate genomic sequence with certain characteristics. For example, [Wilkins et al. \(2024\)](#)  used SpliceAI predictions to create cryptic splice sites, [Schreiber et al. \(2025\)](#)  applied greedy editing of a given sequence by treating each position as a distribution over four nucleotides and computing the gradient per position, and [M Ghari et al. \(2024\)](#)  developed generative flow networks to edit or generate sequences with desired properties. Although these works did not focus on designing sequences for the tissue specific splicing as presented here, such algorithms can be potentially adopted for this task in future works.

There are several potential applications for the work proposed here. First, TrASPr provides a relatively ‘lightweight’ LLM that can be easily fine-tuned for additional cellular conditions of interest. This capability enables the detection of condition-specific splicing and associated regulatory elements in scenarios where experiments have not yet been conducted or in genes with low coverage. It can also be used to assess the effects of genetic variants, such as resolving undiagnosed cases of rare diseases. This application was recently demonstrated in [Wagner et al. \(2023\)](#) , where SpliceAI emerged as a top-performing model. For RNA design tasks, BOS and similar algorithms offer valuable tools for synthetic biology studies and therapeutic applications. For example, these algorithms could guide the design of sequences to target with ASO (antisense oligonucleotide) therapies or prime editing approaches.

While the above applications are exciting, we acknowledge several limitations and areas for potential improvement in this work. First, although TrASPr demonstrated significant advancements in PSI and dPSI predictions, it remains far from perfect. Specifically, the current model was not optimized for predicting the effects of genetic variations and can only capture mutations within 200 bases of existing splice sites. Additionally, we have not evaluated TrASPr on complex splicing variations involving multiple alternative splice junctions. However, we note that such events are often simplified when analyzing changes between two cellular conditions [Vaquero-Garcia et al. \(2023\)](#) .

As for the analyses conducted here, it is important to recognize that the labels used for evaluating prediction tasks are inherently noisy and limited in number. For instance, RNA-Seq quantification is prone to noise, as are RBP binding assays such as eCLIP. Furthermore, RBP regulatory motifs are relatively crude representations, meaning many targets may be missed. Changes observed upon RBP knockdown (KD) could also arise from indirect effects, such as another RBP influenced by the KD, or from other sequence motifs.

Finally, we note a recent trend in AI for genomics toward the development of large ‘foundation’ models that can be fine-tuned for specific tasks. At this time, it remains unclear in what tasks such models can deliver superior performance or when lighter, task-specific models may be more accurate or useful. A related question is how easily a model can be adapted to handle new tasks, conditions, or datasets [Penzar et al. \(2023\)](#) .

In this work, we present a ‘middle ground’: leveraging the framework of LLM pre-training while constructing models tailored to specific tasks. This approach results in significantly lighter models that require fewer computational resources and are easier to adapt to new tasks. Another related approach is to create smaller, dedicated, models that adopt a foundation model latent representation to specific tasks (e.g., [Daoud and Ben-Hur \(2025\)](#) ; [Singh et al. \(2025\)](#) ). We envision TrASPr, paired with BOS for design tasks, as a ‘base’ model that can be readily fine-tuned for specific cell types, tissues, RBP knockdowns, and other conditions. Indeed, we demonstrated TrASPr’s ability to generalize to alternative 3’ and 5’ splice site usage and to unseen cellular conditions by utilizing PCA for a latent space representation of cellular conditions.

In summation, we are excited for the future of LLM for RNA prediction, optimization and design. We hope both the algorithms and the new RNA splicing design task will have a significant impact, serving the community on a variety of current tasks as well as a base for future developments.

Methods

Our method, depicted in Figure 1d, involves two main components: A transformer-based splicing prediction model (TrASPr) and a Bayesian Optimization algorithm (BOS) to design RNA with desired properties. We now turn to describe the two modeling components in order.

TrASPr

Pre-training RNA splice site BERT model

The base model for TrASPr is a 6-layer BERT model pre-trained on human RNA splice sites (Figure 1d). Following the pre-training step, as in Ji et al. (2021), TrASPr takes an RNA sequence converted to 6-mer tokens as input, but instead of using the BERT default maximum length, we feed the model with 400 bases long sequences where the splice site (either 5' or 3' splice site, as shown in the illustration) is in the center.

For pre-training, we follow BERT Devlin et al. (2019) in randomly choosing 15% of tokens but additionally mask the surrounding six tokens for each one to account for our overlapping 6-mer tokenization. We used standard masked autoencoder training, calculating the loss from the original 15% of tokens that were masked. The model is pre-trained for 110k steps with a batch size of 40. The learning rate was set to $4e-4$, and we used a linear scheduler with 10000 warm-up steps.

The TrASPr model and fine-tuning

The structure of TrASPr is depicted in Figure 1d. Each transformer encoder is a 6-layer transformer with 12 heads. We set the hidden size of transformers as 768 for each layer. For any given AS event e , the input to TrASPr is a sequence composed of four sequences $S_e = \{S_e^i\}_{i=1}^4$ such that each S_e^i covers the exonic and intronic regions surrounding one of the four splice sites involved in the exon skipping AS event e . Each S_e^i is fed through a matching pre-trained transformer T^i , which also accepts additional event features $F_e = \{F_{e,i}\}$ (see below). The latent space representation from each transformer T^i , captured by their respective CLS tokens, are concatenated together along with the feature set F_e and fed into two hidden layer MLP with layer widths 3080 and 768.

Event features

The additional feature set F_e includes the exon and intron length information as discrete tokens (binned by length) and the tissue type. We also include discretized conservation generated based on the PhastCons score Siepel et al. (2005) for each k-mer in the sequence. Exons generally have significantly higher conservation values, as these reflect selection pressure due to non-splicing related functions (coding for proteins). We, therefore, used a constant value for all exonic regions, which is the mean of all conservation scores across all intron positions in the training set, and kept the original values for the introns. In addition, we incorporated three types of indicators into the model: *de novo*, coding region, and frame-shift indicators. The *de novo* indicator specifies whether the middle exon is absent from existing annotations. The coding region indicators are 5-digit binary flags to denote whether the middle exon falls within a coding region, a non-coding region, a start-coding region, an end-coding region, or a combination of multiple categories. The frame-shift indicator identifies whether the event results in a frame-shift alternative splicing outcome.

Tissue representation with PCA

Tokens are widely used in language models to represent segments of sequence. However, token-based representations are inherently uninformative without context, and unseen tokens during training cannot be interpreted or utilized by the model. To address these issues, we propose an approach to learn the representation of tissues based on gene expression profiles. Specifically, we construct tissue representations using the expression levels of a curated list of 3,344 RNA-binding

proteins (RBPs). Each tissue vector was constructed with 15 samples from the GTEx dataset, with expression levels measured in TPMs (transcripts per million) obtained from the GTEx website. For our experiments, we selected eight tissues from GTEx (lung, heart, cerebellum, adrenal gland, liver, spleen, cortex, and EBV-transformed lymphocytes) and two cell lines from ENCODE (HepG2 and K562). The resulting representation vectors were then normalized using min-max normalization. Applying principal component analysis (PCA) Wold et al. (1987) to these expression level vectors, we extract the top 50 components as the representation for the corresponding tissue. These PCA-derived representations are then incorporated into the feature vector by concatenating them to the sequence representation produced by the transformer encoders as shown in Figure 1d. The model is trained as described in the experiment section.

Supervision

Since we are interested in learning splicing variations between different conditions we define target variables that force the model to learn those Jha et al. (2017). Specifically, based on the splicing outcome for an event $e \in E$ in two conditions $c, c' \in C$, where E is all the cassette exon events and T is all the tissue or cell line conditions, the target variables include:

$$\begin{aligned}\Psi_{e,c} &= E[\Psi_{e,c}] \\ \Delta\Psi_{+e,c,c'} &= |\max(\epsilon, E[\Delta\Psi_{e,c,c'}])| \\ \Delta\Psi_{-e,c,c'} &= |\min(\epsilon, E[\Delta\Psi_{e,c,c'}])|\end{aligned}$$

Here $E[\Psi_{e,c}]$, $E[\Delta\Psi_{e,c,c'}]$ represent the posterior expected values for PSI and Delta PSI(dPSI) as estimated by MAJIQ from the RNA-Seq experiments Vaquero-Garcia et al. (2016). The $\Delta\Psi_{+e,c,c'}$ target captures events with increased inclusion level between tissue c and c' while $\Delta\Psi_{-e,c,c'}$ captures events with increased exclusion, incentivizing the model to focus its attention on splicing changes.

To avoid the zero gradient issue, we use a random small number between 0.001 and 0.002 as ϵ .

$$Loss = \sum_{e \in E, c, c' \in C} [\mathcal{L}(\Psi_{e,c}, \hat{\Psi}_{e,c}) + \mathcal{L}(\Delta\Psi_{+e,c,c'}, \Delta\hat{\Psi}_{+e,c,c'}) + \mathcal{L}(\Delta\Psi_{-e,c,c'}, \Delta\hat{\Psi}_{-e,c,c'})]$$

As shown above, we use the cross-entropy as the loss function L , which performed better than the mean-squared error loss during our ablation studies. In the fine-tuning step, we train the model with a $2e-5$ learning rate and batch size of 32 for ten epochs.

Sequence design for splicing outcomes

Beyond supervised learning, we also demonstrate that TrASPr can be leveraged to solve sequence design problems. Given a genomic sequence context $S_e = (s_1, \dots, s_n)$, made of a cassette exon e and flanking intronic/exonic regions, TrASPr predicts for tissue c the fraction of transcripts where exon e is included or skipped over, $\Psi_c(S_e)$. This inclusion value can directly be used as the basis for optimization problems, where we seek to design new sequences $\tilde{S}_e$ that differ from S_e only slightly but exhibit altered splicing outcomes. Formally, we define these optimization problems as:

$$\begin{aligned}\arg \min_{\tilde{S}_e} \Psi_c(\tilde{S}_e) \text{ s.t. } \text{lev}(\tilde{S}_e, S_e) \leq \tau \\ \text{or } \arg \max_{\tilde{S}_e} \Psi_c(\tilde{S}_e) \text{ s.t. } \text{lev}(\tilde{S}_e, S_e) \leq \tau\end{aligned} \quad (1)$$

Here, $\text{lev}(\tilde{S}_e, S_e)$ denotes the Levenshtein distance between $\tilde{S}_e$ and S_e . Solving the minimization problem is equivalent to finding a small perturbation (up to edit distance τ) of S_e that *reduces* inclusion in the target tissue c by as much as possible. The maximization problem corresponds to *increasing* inclusion. In practice, we add additional constraints that $\forall c' \neq c$ and $\Psi_{c'}(\tilde{S}_e)$ cannot be reduced below 0.05. These additional constraints prevent an optimization routine from destroying splicing to such an extent that all inclusion levels are driven to zero.

To solve this optimization problem, we adapt recent work in latent space Bayesian optimization (LSBO) for black-box optimization problems over structured and discrete inputs Maus et al. (2022); Stanton et al. (2022); Gligorijević et al. (2021); Moss et al. (2020); Winter et al. (2019); Sanchez-Lengeling and Aspuru-Guzik (2018); Gómez-Bombarelli et al. (2018); Griffiths

and Hernández-Lobato (2020) [↗](#); Grosnit et al. (2021) [↗](#). LSBO solves structured optimization problems using two primary components: (1) a deep variational autoencoder (VAE) model and (2) a Bayesian optimization routine.

Variational autoencoders for LSBO

In LSBO, we train a VAE that assists in reducing the discrete optimization problem over sequences S to a continuous optimization problem over the *latent space* of the VAE, $\mathcal{Z} \subset \mathbb{R}^d$. Leveraging the same data used to train TrASPr, we train a 6 layer transformer encoder $\Phi: S \rightarrow \mathcal{P}(\mathcal{Z})$ and 6 layer transformer *decoder* $\Gamma: \mathcal{Z} \rightarrow \mathcal{P}(S)$ Vaswani et al. (2017) [↗](#). Here, $\mathcal{P}(\mathcal{Z})$ denotes the probability over the latent space $\mathcal{Z} \subset \mathbb{R}^d$, and $\mathcal{P}(S)$ denotes the probability over the sequence space S . The encoder $\Phi(S_e)$ maps sequences S_e onto a distribution over real-valued, 256-dimensional, continuous latent vectors $\mathbf{z} \in \mathbb{R}^{256}$. The decoder $\Gamma(\mathbf{z})$ reverses this process probabilistically. The parameters Φ and Γ are trained so that we have $\Gamma(\Phi(S_e)) \approx S_e$. Because we only care about the output sequence $\tilde{S}_e$, here we abuse notation and denote the most probable sequence output from the decoder as $\Gamma(\mathbf{z})$. For optimization, the advantage the VAE provides is the ability to optimize over *latent vectors* $\mathbf{z} \in \mathbb{R}^{256}$ rather than directly over sequences S_e . This is because, for any $\mathbf{z}$ proposed by an optimization algorithm, we can evaluate $\Psi_c(\Gamma(\mathbf{z}))$. We therefore search for a $\tilde{\mathbf{z}}$ such that $\tilde{S}_e := \Gamma(\tilde{\mathbf{z}})$ is an optimal solution to the optimization problem.

Bayesian optimization

With the optimization problem in Equation 1 [↗](#) reduced to a continuous problem over $\tilde{\mathbf{z}} \in \mathcal{Z}$, we can now apply standard continuous black-box optimization algorithms. Bayesian optimization Garnett (2023) [↗](#) is among the most well-studied of these approaches in the machine learning literature. In iteration n of Bayesian optimization, we have a dataset $\mathcal{D}_n = \{(\mathbf{z}_i, y_i)\}_{i=1}^n$ for which $y_i = \Psi_c(\Gamma(\mathbf{z}_i))$ is the known objective value. We train a surrogate model of the objective function using this data—most commonly a Gaussian process Rasmussen (2003) [↗](#)—and use this surrogate to inform a policy—commonly called an *acquisition function*—that determines what latent vectors $\mathbf{z}_{n+1}$ to consider next. Here, we used <monospace>LOL-BO</monospace> Maus et al. (2022) [↗](#) as our base LS-BO algorithm. To accommodate the constraints in Equation 1 [↗](#), we modify LOL-BO to utilize SCBO Eriksson and Poloczek (2021) [↗](#) rather than TuRBO Eriksson et al. (2019) [↗](#) as the underlying optimization routine. As with the objective, the Levenshtein constraint is evaluated on the decoded latent vectors: $\text{lev}_Z(\mathbf{z}, \mathbf{z}') = \text{lev}(\Gamma(\mathbf{z}), \Gamma(\mathbf{z}'))$.

Experimental validation

Re-Sequencing Low-Coverage Cassette Events with LSV-Seq

LSV-Seq was used to generate targeted libraries for this list of low-coverage cassette events prioritized for resequencing with TrASPr Yang et al. (2024) [↗](#). Briefly, primers for LSV-Seq were designed using the Optimal Prime algorithm, which uses machine-learning models to optimize primer sequences for both specificity and yield. To perform LSV-Seq, the resulting primers were synthesized as a single combined pool of over 1,000 primers and used in the first-strand reverse transcription reaction for RNA from each tissue of interest. Sequenceable libraries were created after additional reactions including second-strand synthesis, in vitro transcription, fragmentation, secondary reverse transcription, and final PCR amplification. The resulting libraries were aligned with STAR (Dobin et al. (2013) [↗](#)). Quantification and visualization of psi values across tissues was performed with MAJIQ/VOILA (Vaquero-Garcia et al. (2016) [↗](#), 2023))

Targeting with dCas13d

In order to test regulatory elements predicted by TrASPr we first trained it to predict differential splicing between the two cell lines: SH-SY5Y and HEK-293T. Selecting cassette exons with high confidence predictions for cell-line specific splicing changes we then selected regions for experimental testing using the following procedure. We randomly mutated sequences in the

alternative exon and flanking introns using a 6 bp sliding window. Each region in a window was mutated 5 times to avoid introducing new motifs and the average predicted PSI was compared to the wild type. Finally, we selected top target regions based on predicted dPSI.

For experimental validation, predicted regulatory or negative control sequences were targeted with dCas13d. HEK-293T cells were cultured in DMEM (Gibco, 10569010) supplemented with 10% (v/v) FBS (Thermo Scientific, A3160502) and 1x penicillin-streptomycin (Thermo Scientific, 15140122). For targeting, cells were co-transfected with vectors expressing dCas13d (pXR002, Addgene #109050) and guide RNA (cloned into a custom expression vector) using the CalPhos Mammalian Transfection Kit (Takara, 631312). Cells were collected two days later and RNA was extracted using the Direct-zol RNA Purification Kit (Zymo Research, D2052). RNA was converted to cDNA with the LunaScript RT SuperMix (NEB, M3010L) and PCR for the splicing event was performed with the Q5 Hot Start High-Fidelity 2X Master Mix (NEB, M0494S). PCR reactions were then visualized on a 2% agarose gel in 1X lithium boric acid buffer (Faster Better Media, LB10-1), stained with SYBR Safe (Invitrogen, S33102).

Data

Splice sites and alternative splicing events detection from GTEx

To pre-train the basic BERT RNA model, we first extract 1.5 million 400 bases long sequences around splice sites from the GENCODE v40 human pre-mRNA transcripts database. These splice junctions were extracted directly from the hg38 annotation regardless of whether those junctions are considered alternative or constitutive or their tissue specificity.

For tissue specific splicing quantification we used the GTEx dataset Consortium (2020) [from](#) which we selected six representative human tissues (heart-atrial appendage, cerebellum, lung, liver, spleen, and EBV-transformed lymphocytes). This RNA-Seq data was first aligned using STAR [Dobin et al. \(2013\)](#) with default parameters. Next, we processed the resulting BAM files with MAJIQ to detect and quantify AS events. To construct a high-confidence set of cassette splicing events, we first built a splice graph for the six GTEx tissue groups using MAJIQ [Vaquero-Garcia et al. \(2023\)](#), where we had 15 samples for each tissue group and filtered junctions existing in <20% samples. Then we run MAJIQ dPSI and heterogeneity quantifier between 15 combinations of tissues to capture differential splicing patterns. From these outputs, VOILA modulizer was subsequently applied to classify cassette events into confident changing (dPSI ≥ 0.15 of heterogen quantified) and non-changing (dPSI < 0.08 with confidence > 0.7 of deltapsi quantified) categories. The resulting sets were merged and further refined through a post-processing step that normalized PSI values, removed low-confidence or artifactual events, and retained only those supported across at least two tissues. Finally, sequence-based features were extracted with standardized exon region lengths, ensuring consistent input representations. To enrich the training data with negative examples, “fake” skip junctions were added to constitutive exon events and small random PSI values were assigned.

For AS events involving alternative 3' and 5' splice sites, we used the following procedure to generate a set of events to train and test on. For each alternative splice site type, we first scan for such alternative splice sites from the MAJIQ splice graphs for the analyzed tissues. If no qualified alternative splice site for the chosen tissues exists, other real alternative ones that are expressed in other tissues will be selected based on Ensembl annotation and MAJIQlopedia. These non-expressed alternative splice sites are assigned with a small PSI value ~ 0.01 . In the end, we got 1883 events for 5' and 1861 events for 3' splice site dataset.

AS event filtering to avoid information leakage

When training and testing on AS events data, care must be taken to avoid testing on similar events that result in information leakage. This is especially important for large models that can easily memorize genomic sequences [Schreiber et al. \(2020\)](#). One source of event similarity are Paralog genes. Paralogs, derived from a common ancestral gene, may result in similar AS events exhibiting similar splicing patterns. To avoid this potential information leakage hazard, we first hide chromosomes (chr1,3,5,7,9 for experiments involved with Pangolin as in [Zeng and Li \(2022\)](#) and

8, 14 for the rest of experiments) for testing, then discarding test events similar to our training set. In addition, similar events in the training or test events can also occur due to overlapping cassette events detection. Specifically, cassette exons extracted from MAJIQ's splice graphs may overlap (e.g., different splice sites used to define the skipped exon). This may be useful for training on diverse exon/intron definitions, but care must be taken to avoid information leakage to the test data. To remove similar AS events due to either paralogs or overlapping AS event detection we follow the procedure used in Jha et al. (2017) [\[17\]](#). Specifically, exon sequence similarity was assessed using BLAT Kent (2002) [\[18\]](#) with filters to identify the difference in length and the estimated similarity p-value. (BLAT settings `maxLenDiff=5`, `minPval=0.0001` and `minIdentity=95`). These settings account also for short exons with high similarity but enough divergence relative to their short length not to achieve a significant p-value.

AS events quantification

Feature generation for 3' and 5' alternative splice site events followed processing steps similar to the ones used for cassette exon events, with the same filter parameters. Similar to the “fake” skip junctions over constitutive exons we also added here AS events where one of the two alternative splice-site junctions was not actually observed in the six GTEx tissues but was nonetheless annotated. For such junctions we first searched if they existed in the original data, then searched the Ensembl annotation D.B., then searched the MAJIQlopedia Quesnel-Vallières et al. (2024) [\[19\]](#) D.B. to see if they were detected across GTEx. Finally, if no such junction was found we introduced a predicted “fake” junction by selecting the best scoring position 400 up/downstream from the existing junction using SpliceAI.

We measure splicing across $c \in [1, \dots, C]$ conditions for events $e \in [1, \dots, E]$. Each AS event e has a sequence S_e comprised of 4 different regions, each centered around the respective splice site $S_e = \{S_e^1, S_e^2, S_e^3, S_e^4\}$. Similarly, each event has a set of associated features, such as exon length, conservation etc., denoted F_e . Splicing quantification for event e in condition c is denoted $\Psi_{e,c} \in [0, 1]$ and differential splicing as $\Delta\Psi_{e,c,c'} \in [-1, 1]$ accordingly. However, we frequently drop the event e or condition c index for brevity.

The above pipeline results in AS event quantification which can be directly fed as input for TrASPr. However, Pangolin and SpliceAI models do not define specific splicing events such as cassette exons. Instead, they use a 10Kb sequence window and predict if the center position is a 3' or 5' splice site. In addition, Pangolin predicts the ‘splice usage’ value for that genomic position, where the usage for a given genomic position in a given tissue is computed from RNA-Seq using SpliSER Dent et al. (2021) [\[20\]](#) during Pangolin's training. To make Pangolin and SpliceAI comparable for predicting AS event inclusion levels, we input the algorithms with 3' and 5' splice site of each alternative exon e . We tested different options to translate the splice sites scores by these algorithms to PSI and found the best results were achieved by averaging the predictions of both 3' and 5' splice site usage. We also tried retraining Pangolin with actual PSI values from MAJIQ with similar results (see main text).

We used both the cassette exon and alternative splice site datasets to evaluate the ability of TrASPr to model tissue-specific splicing. To ensure robust assessment, we removed paralogous and overlapping events through chromosome masking and sequence similarity filtering, resulting in independent datasets for training TrASPr and benchmarking against existing models. AS event data is collected in `alternative_3.tsv` and `alternative_5.tsv`.

Splice site swap and disruption for RNA sequence design

The splice site plays a critical role in alternative splicing. Identifying and then restoring them represents a crucial milestone for TrASPr and BOS. To assess how TrASPr responds to modifications in splice site strength, we scored all donor and acceptor splice sites using MaxEnt Yeo and Burge (2003) [\[21\]](#). Based on these scores, we classified splice sites as strong or weak by selecting the top 15% (5' score > 10.17, 3' score > 10.83) and bottom 15% (5' score < 5.95, 3' score < 5.52) from the distribution of splice site scores in the test set. For low inclusion sequences (PSI < 0.15), we substituted their 3' and 5' splice sites in the middle exon with strong splice sites.

Similarly, for high-inclusivity sequences ($\text{PSI} > 0.85$), we replaced the existing splice sites with weaker ones. Our goal was to observe TrASPr's predicted changes in PSI before and after each replacement.

After confirming TrASPr's ability on weak and strong splice site recognition, we further designed the experiment to test if BOS is able to recover the disrupted splice sites. In this case, we followed the same strategy to select high-inclusive sequences and strong splice sites. These strong 3' or 5' splice sites were disrupted through random mutations. Then, we randomly selected eight sequences each for the 3' and 5' splice sites as starters for generative models aimed at increasing PSI. In the end, the successfully generated sequences will be scored by MaxEnt to determine if their PSI is increased by generating strong splice sites.

Mutations and knockdown data

To evaluate the capability of TrASPr and BOS to predict or suggest mutations, we curated four other datasets. The first dataset is the ENCODE database for RBP Knockdown (KD) in human cell lines Van Nostrand et al. (2020) [\[1\]](#), where we focused on three well-studied RBPs (TIA1, PTBP1, QKI). This data resulted in a list of 59 putative RBP regulatory targets for which we could 'remove' the effect of these RBPs on the set of their AS targets by randomly mutating their identified binding motifs. This set was then supplemented with nine validated targets of RBFOX from a recent study Maurin et al. (2023) [\[2\]](#). In the prediction results, "change" and "no change" are determined based on a threshold set at the 95th percentile of observed effects from random mutations. To mitigate the positional effect of mutations, we utilized the same relative distance of the RBP binding sites from splice sites in our original set. For each position, we randomly select 100 events from the test set and performed five different random mutations. To ensure a fair comparison across different models, we evaluated all models on the same random mutation dataset and calculated their 95th percentile thresholds, which correspond to the following values. TrASPr:

$\Delta\text{PSI} > 0.019$, SpliceAI: $\Delta\text{Usage} > 0.043$, Pangolin: $\Delta\text{Usage} > 0.047$. We note that these threshold values are specific to the exons and RBP sites we tested, aimed to create a uniform testing procedure for all methods. Finally, we included two additional datasets to assess splicing outcome predictions in the presence of genetic mutations. One is a recent high-throughput assay with 6106 mutation combinations around exon2 from the CD19 gene Cortés-López et al. (2022) [\[3\]](#). This data is typical of assays that measure many mutations in a clinically relevant event, not necessarily tissue-specific ones. For capturing a tissue-specific event, we also included low-throughput experiments from a mini-gene reporter assay where the effect of mutating several regions upstream of the neuronal-specific exon 16 of the mouse Daam1 gene was tested Barash et al. (2010) [\[4\]](#).

Data availability

RNA-Seq data used to train the models is available from GTEx repository dbGAP phs000424.v8.p2. LSV-Seq experimental data are available on GEO under accession number GSE246294 Trained model, code, and data to generate figures are available <https://bitbucket.org/biociaphers/traspr/src/main/> [\[5\]](#)

Supplementary figures

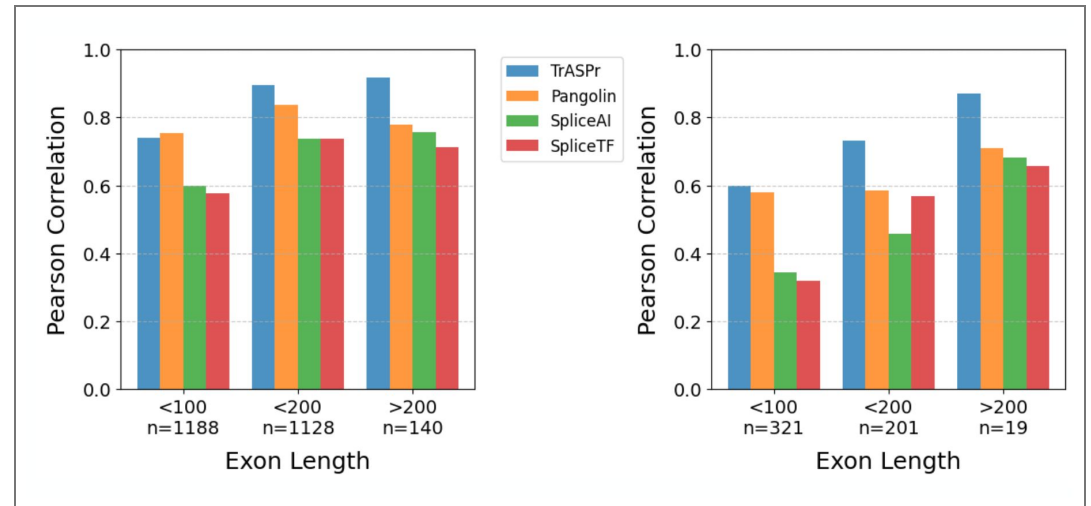


Figure 2 - Supp 1. GTEx Test set Pearson correlation for each model, binned by the combined length of the cassette exon's upstream and downstream introns. Left: All test cases. Right: Only for samples involving a splicing change (dPSI ≥ 0.15)

Figure 2 - Supp 2. Same as Figure 2 - Supp 1 but here test cases are binned by the alternative exon's length.

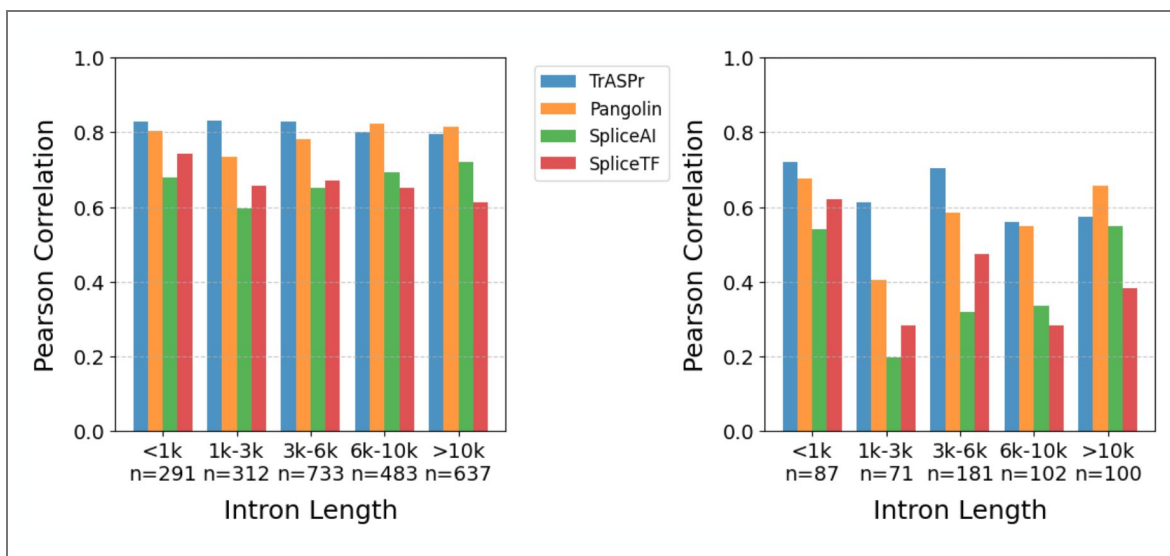
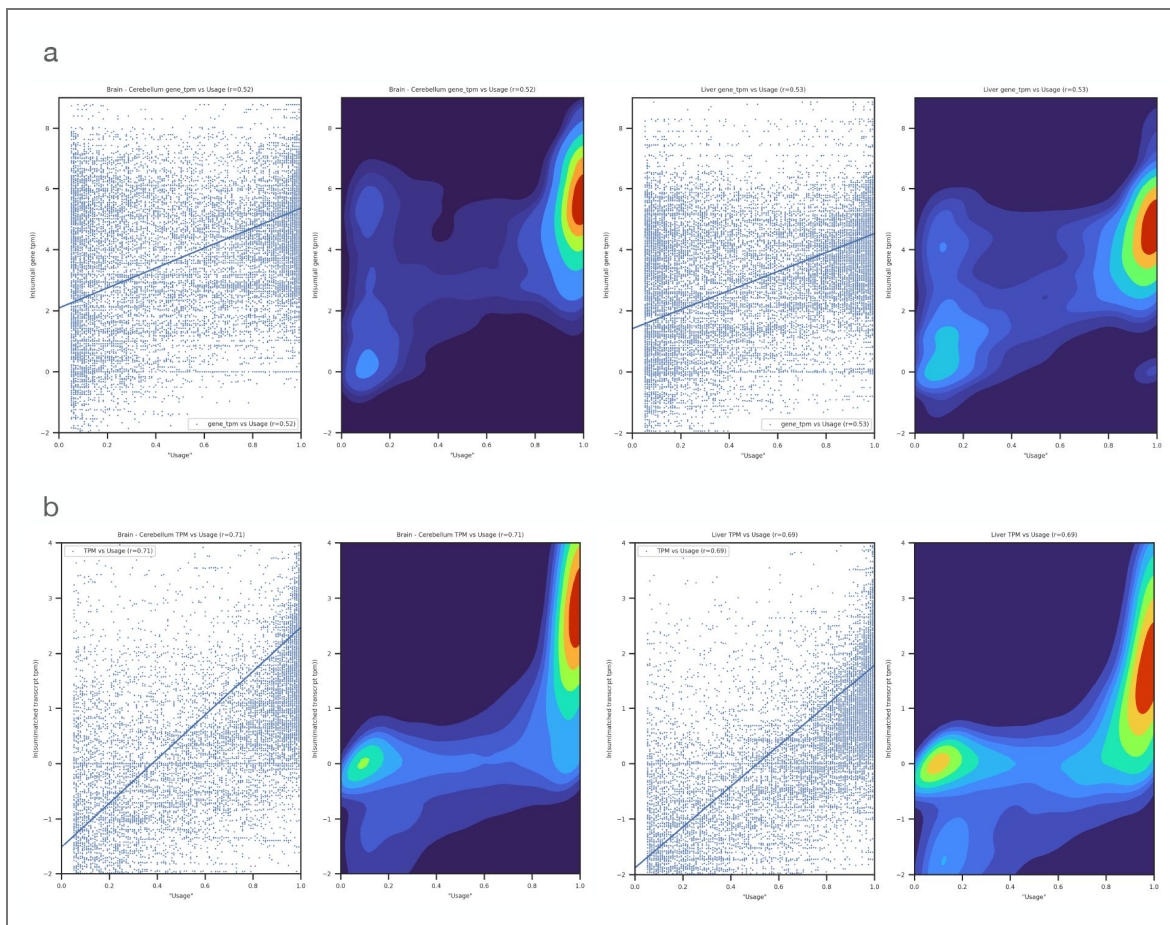


Figure 2 - Supp 3.

(a) Pearson correlation between gene expression (TPM) and splice site usage as defined by SpliceTransformer using GTEx Brain Cerebellum ($r = 0.52$, left) and liver ($r = 0.53$, right). (b) Correlation is further improved when considering the expression of only the isoforms that contain a specific splice site (cerebellum $r = 0.71$, liver $r = 0.69$ right).



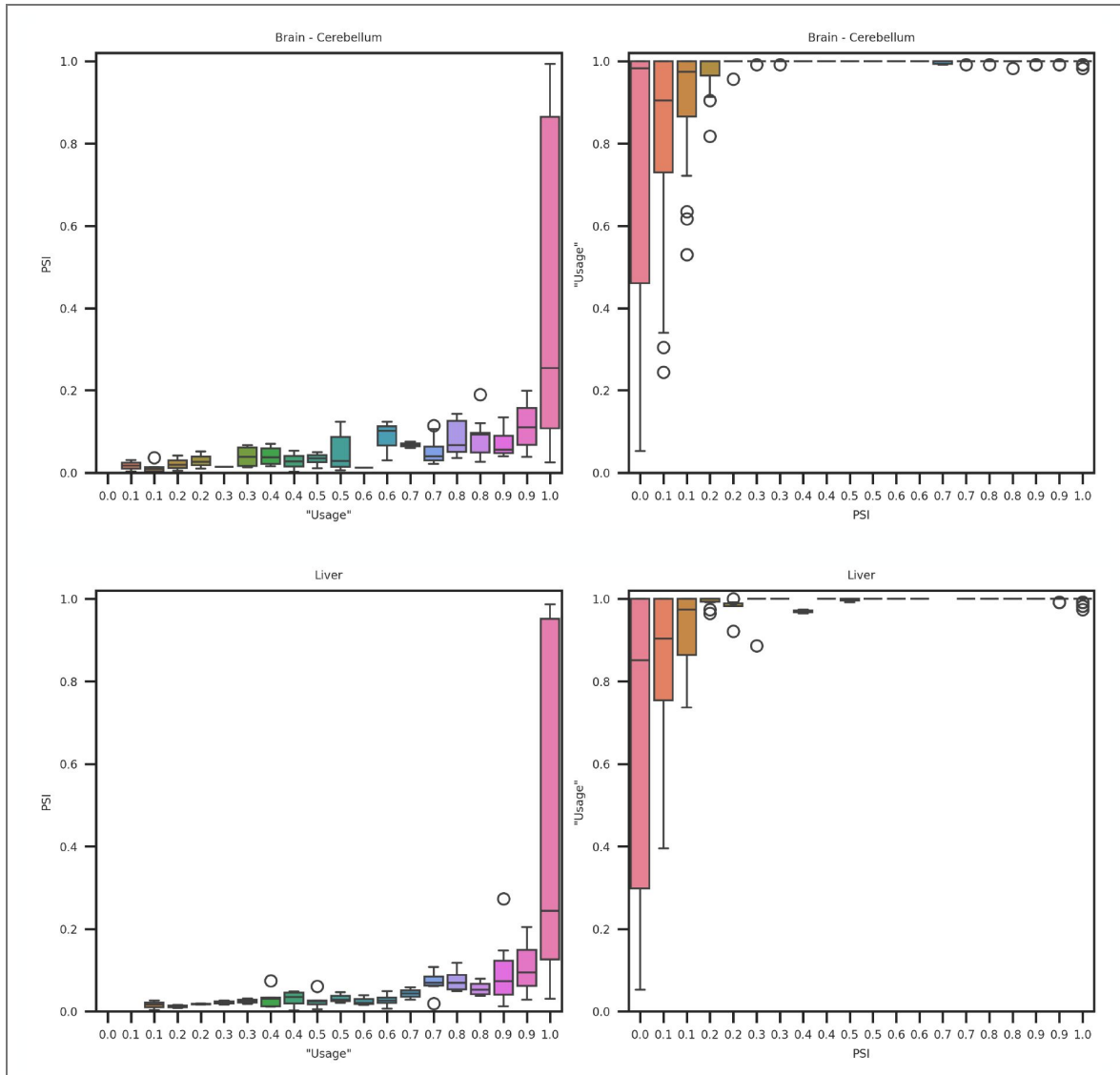


Figure 2 - Supp 4. Tissue PSI values in GTEx samples (top - Cerebellum, bottom - liver) vs. SpliceTransformer usage.

Correlation between usage and PSI is weak (Pearson $r = 0.076$ for Cerebellum, $r = 0.074$ for liver). Only shown are cassette exon PSI in chromosome 8 which were quantified by MAJIQ with high confidence and used as test data for all algorithms in the main text Fig2. When usage is high (left panels) PSI has the typical bimodal distribution such that it can either be very high or very low. Conversely, when PSI is low coverage can greatly vary between 0.05 (the threshold filter set by the SpliceTransformer authors) and 1 but when PSI is high the events are detected in almost all samples (usage close to 1).

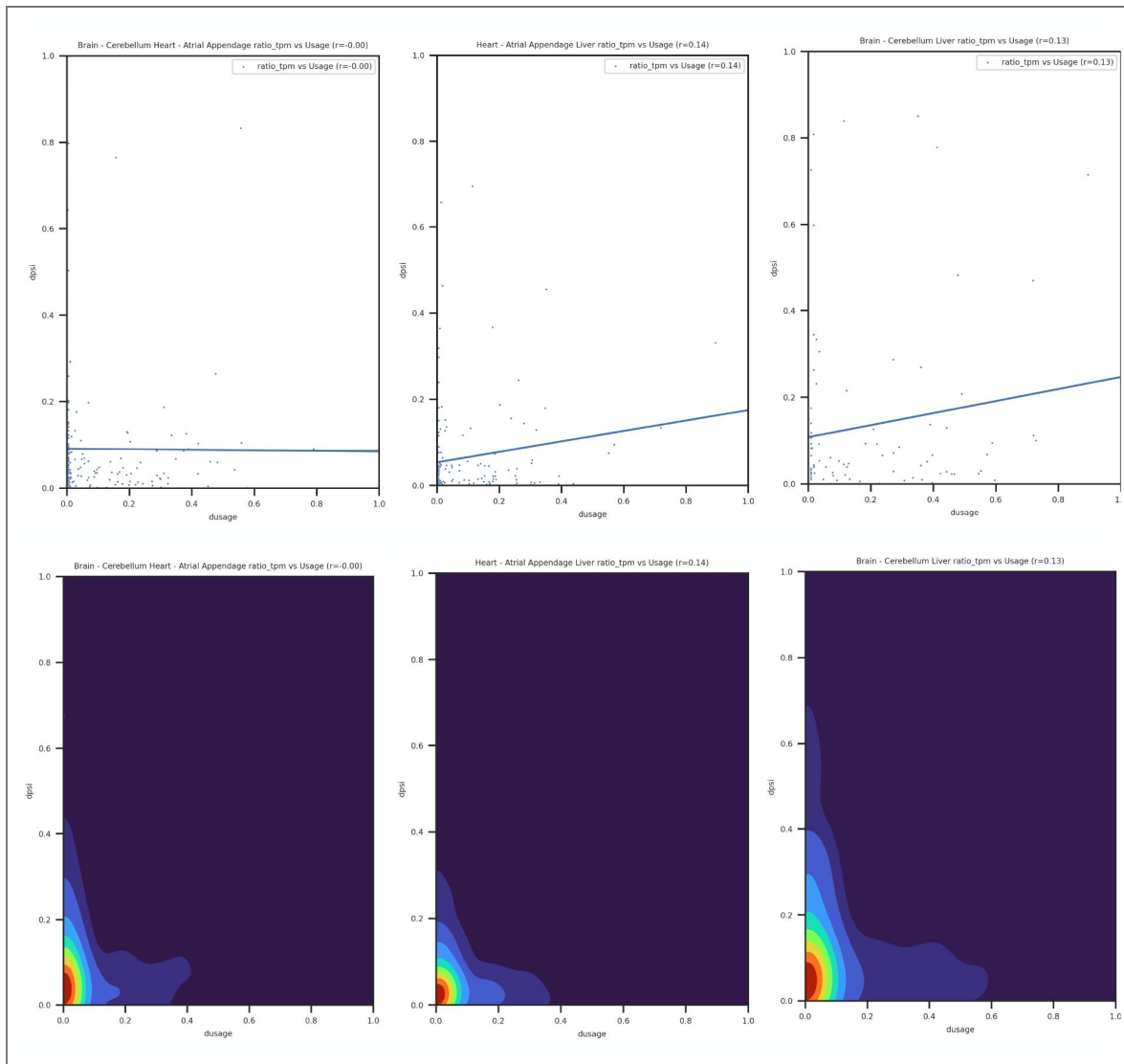


Figure 2 - Supp 5. Differential usage (x-axis) vs. differential splicing (dPSI, y-axis) for cassette exons in chromosomes 7,8 assessed for the three GTEx tissue pairs used in the main text (Fig 2): heart_BCer, heart_liver, BCer_liver.

dPSI values are the same as those used to train and test all algorithms in the main text. Top: Scatter plot Bottom: matching heat map. Note that ~ 80-90% of the samples with dPSI > 0.1 have dUsage ~ 0 and therefore will not be captured by the SpliceTransformer target function which weighs samples by their dUsage. A few points exhibit high dUsage and high dPSI contributing to some correlation, with pearson correlation ranging from 0 to 0.14 depending on the tissue pair.

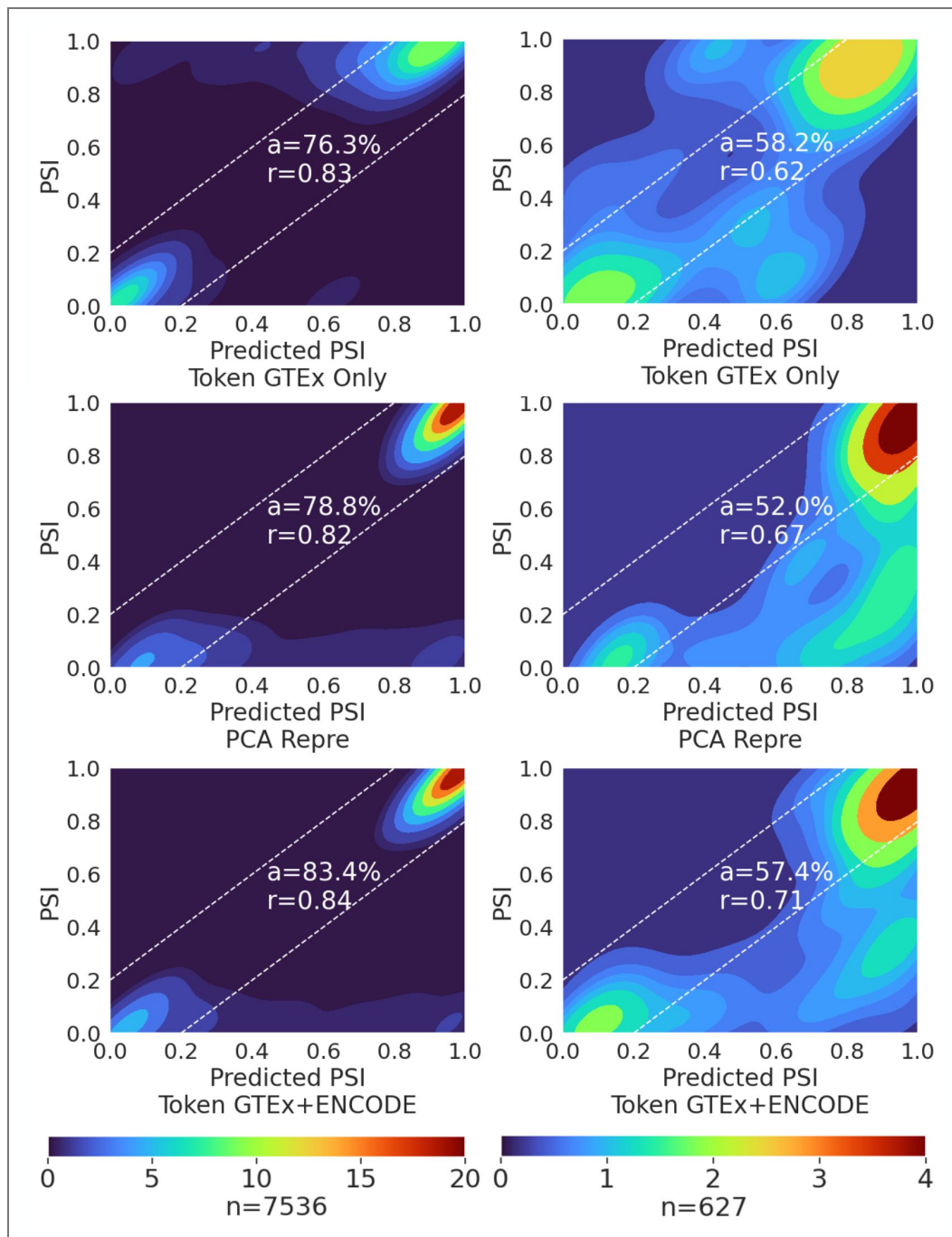


Figure 3 - Supp 1. TrASPr was trained on GTEx 6 tissues or GTEx + two ENCODE cell lines and then tested on two cell lines in ENCODE (HepG2, K562).

Top: The tissues were first represented as tokens, and new cell line results were predicted based on the average over conditions during training. Mid: TrASPr used the PCA learned representation to predict AS in the two ENCODE cell lines it never trained on. Bottom: TrASPr was trained with token represented tissues on GTEx + two ENCODE cell lines. Left: All samples. Right: Changing events

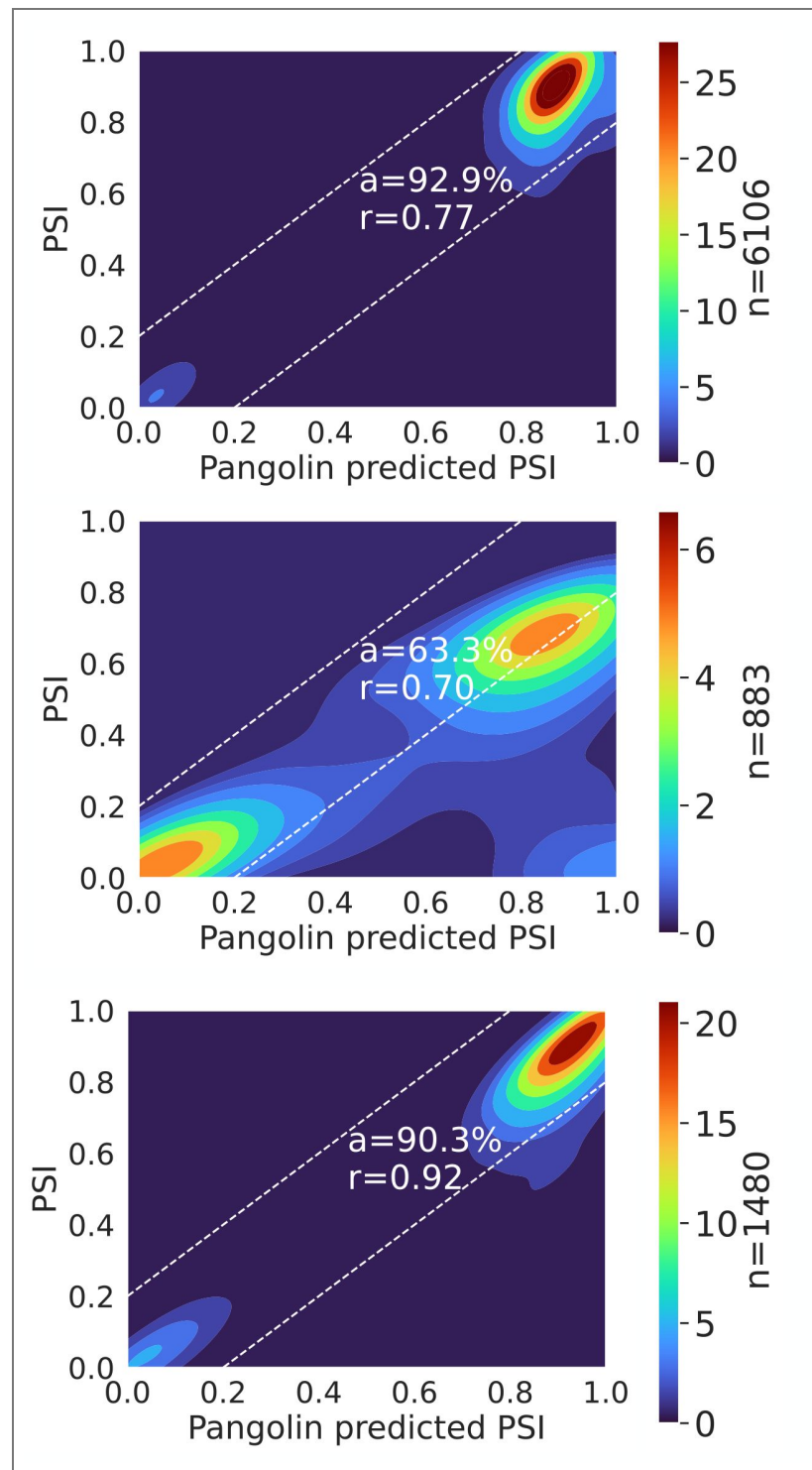


Figure 4 - Supp 1. Heatmaps showing the performance of Pangolin in predicting the effect of mutations for CD19 dataset with the same settings as shown in Figure 4c [DOI](#)

Figure 4 - Supp 2. CDF for the difference between TrASPr test data predictions (PSI') and ground truth PSI derived from MAJIQ on ENCODE dataset.

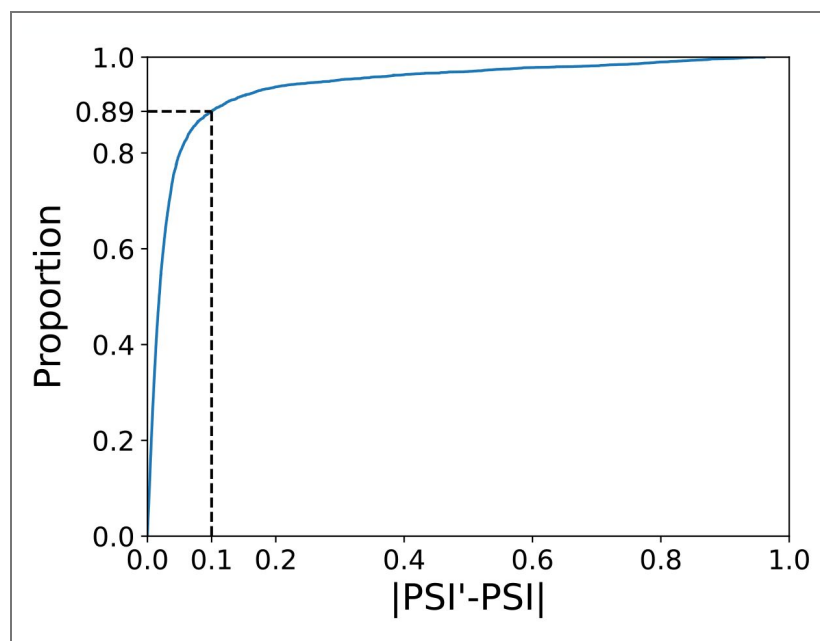


Figure 4 - Supp 3. Predicting the effect of tissue specific splice factors exhibits strong positional biases.

The sequence motif of RBFOX (TGCAATG - left) and QKI (ACTAAC - right) was inserted at each of the modeled positions around 80 cassette exons, 40 of which are highly included (PSI>0.75) and 40 are lowly included (PSI<0.25). Predictions were compared to 10 random kmers inserted in the same location (see Methods). Colors represent the indicated percentile, per position, of effects ranging from the 10th percentile, where increased exclusion is observed over the 80 cassette events, to the 95th percentile, where increased inclusion is observed.

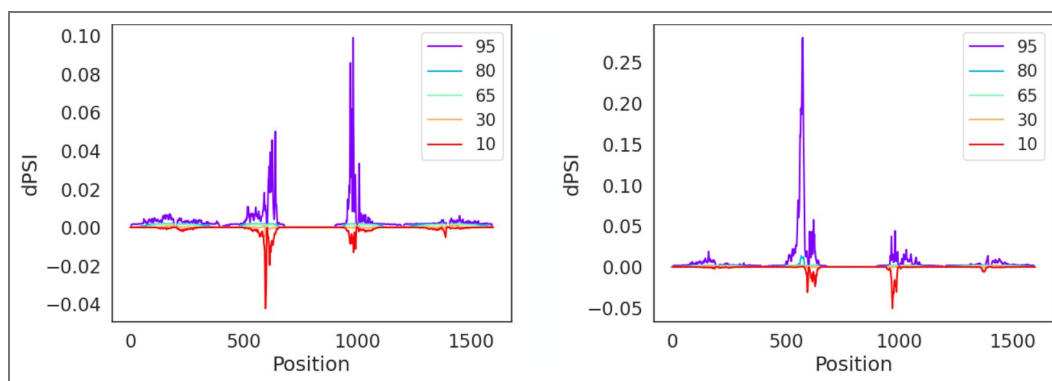


Figure 6 - Supp 1. BOS sequence edits for neuronal specific splicing.

This figure matches Figure 6C but includes all BOS generated samples that pass the user-defined criteria of brain-specific dPSI > 0.2 and not just the top 20th percentile for the best generated samples. Left: The overall number of edits by BOS per position over all generated sequences (n=20349). Right: A CDF of TrASPr based predicted neural specific dPSI after the BOS edits (blue), and when intronic edits that are further than 8bp from the nearest splice sites are removed (orange). The shift between the blue and orange curves demonstrates that almost all of the predicted neuronal specific effect is achieved via intronic edits. We further tested the effect of removing edits only from the downstream (green) and upstream (red) intronic region flanking the middle exon.

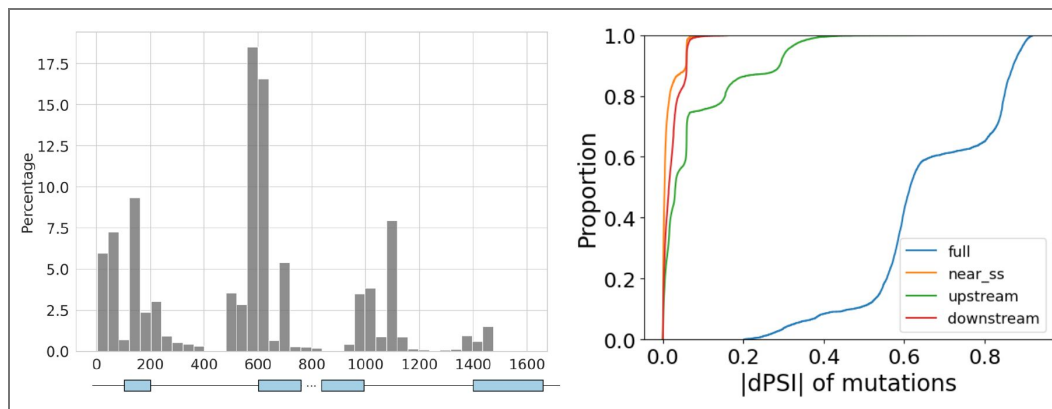
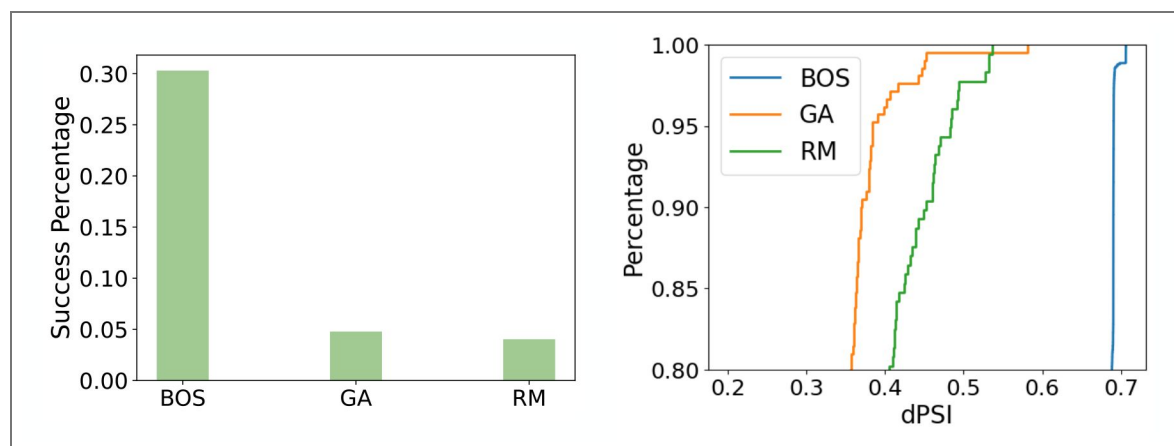


Figure 6 - Supp 2. Generated sequences result for Daam1 gene exon 16 (n=4392).

Algorithms were tasked with reducing inclusion in N2A cell line (dPSI>0.2) but not destroying the inclusion in other tissues(PSI>0.1). Left: success rate of each algorithm to indicate the efficiency. Right: CDF of dPSIs for each algorithm to indicate the effectiveness.



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Additional information

Author Contributions

YB conceived the project. DW, and YB developed TrASPr and planned the experiments, with input from AJ. DW performed all the analysis. AJ developed earlier non-Transformer versions of the model and together with SJ developed all the data processing pipelines. NM and JRG developed BOS with input from DW and YB. NM with DW support performed BOS experiments and DW performed BOS results analysis. PC directed LSV-Seq and dCas13 experiments performed by KY, BW-M, and AT. KY analyzed LSV-Seq results and BW-M performed LSV-Seq case analysis.

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Author ORCID iDs

Anupama Jha: <https://orcid.org/0000-0003-3029-2086>

Peter Choi: <https://orcid.org/0000-0002-2820-3032>

Yoseph Barash: <https://orcid.org/0000-0003-3005-5048>

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Peer reviews

Reviewer #1 (Public review):

Summary

The authors propose a transformer-based model for prediction of condition- or tissue-specific alternative splicing and demonstrate its utility in design of RNAs with desired splicing outcomes, which is a novel application. The model is compared to relevant existing approaches (Pangolin and SpliceAI) and the authors clearly demonstrate its advantage. Overall, a compelling method that is well thought out and evaluated.

Strengths:

- (1) The model is well thought out: rather than modeling a cassette exon using a single generic deep learning model as has been done e.g. in SpliceAI and related work, the authors propose a modular architecture that focuses on different regions around a potential exon skipping event, which enables the model to learn representations that are specific to those regions. Because each component in the model focuses on a fixed length short sequence segment, the model can learn position-specific features. Furthermore, the architecture of the model is designed to model alternative splicing events, whereas Pangolin and SpliceAI are focused on modeling individual splice junctions, which is an easier problem.
- (2) The model is evaluated in a rigorous way - it is compared to the most relevant state-of-the-art models, uses machine learning best practices, and an ablation study demonstrates the contribution of each component of the architecture.
- (3) Experimental work supports the computational predictions: Regulatory elements predicted by the model were experimentally verified; novel tissue-specific cassette exons were verified by LSV-seq.

(4) The authors use their model for sequence design to optimize splicing outcome, which is a novel application.

Weaknesses:

None noted.

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Reviewer #2 (Public review):

Summary:

The authors present a transformer-based model, TrASPr, for the task of tissue-specific splicing prediction (with experiments primarily focused on the case of cassette exon inclusion) as well as an optimization framework (BOS) for the task of designing RNA sequences for desired splicing outcomes.

For the first task, the main methodological contribution is to train four transformer-based models on the 400bp regions surrounding each splice site, the rationale being that this is where most splicing regulatory information is. In contrast, previous work trained one model on a long genomic region. This new design should help the model capture more easily interactions between splice sites. It should also help in cases of very long introns, which are relatively common in the human genome.

TrASPr's performance is evaluated in comparison to previous models (SpliceAI, Pangolin, and SpliceTransformer) on numerous tasks including splicing predictions on GTEx tissues, ENCODE cell lines, RBP KD data, and mutagenesis data. The scope of these evaluations is ambitious; however, significant details on most of the analyses are missing, making it difficult to evaluate the strength of evidence.

In the second task, the authors combine Latent Space Bayesian Optimization (LSBO) with a Transformer-based variational auto encoder to optimize RNA sequences for a given splicing-related objective function. This method (BOS) appears to be a novel application of LSBO, with promising results on several computational evaluations and the potential to be impactful on sequence design for both splicing-related objectives and other tasks. However, comparison of BOS against existing methods for sequence design is lacking.

Strengths:

- A novel machine learning model for an important problem in RNA biology with excellent prediction accuracy.

- Instead of being based on a generic design as in previous work, the proposed model incorporates biological domain knowledge (that regulatory information is concentrated around splice sites). This way of using inductive bias can be important to future work on other sequence-based prediction tasks.

Weaknesses:

- Most of the analyses presented in the manuscript are described in broad strokes and are often confusing. As a result, it is difficult to assess the significance of the contribution.

- As more and more models are being proposed for splicing prediction (SpliceAI, Pangolin, SpliceTransformer, TrASPr), there is a need for establishing standard benchmarks, similar to those in computer vision (ImageNet). Without such benchmarks, it is exceedingly difficult to compare models.

*This point is now addressed in the revision *

*Moreover, datasets have been made available by the authors on BitBucket. *

- Related to the previous point, as discussed in the manuscript, SpliceAI and Pangolin are not designed to predict PSI of cassette exons. Instead, they assign a "splice site probability" to each nucleotide. Converting this to a PSI prediction is not obvious, and the method chosen by the authors (averaging the two probabilities (?)) is likely not optimal. It would be interesting to see what happens if an MLP is used on top of the four predictions (or the outputs of the top layers) from SpliceAI/Pangolin. This could also indicate where the improvement in TrASPr comes from: is it because TrASPr combines information from all four splice sites? Also consider fine-tuning Pangolin on cassette exons only (as you do for your model).

*This point is still not addressed in the revision. *

- L141, "TrASPr can handle cassette exons spanning a wide range of window sizes from 181 to 329,227 bases-thanks to its multi-transformer architecture." This is reported to be one of the primary advantages compared to existing models. Additional analysis should be included on how TrASPr performs across varying exon and intron sizes, with comparison to SpliceAI, etc.

Added after revision: The authors have added additional analyses of performance based on both the length of the exon under consideration and the total length of the surrounding intronic contexts. The result that TrASPr performs well across various context sizes (i.e., the length of the sequence between the upstream and downstream exons, ranging from <1k to >10k) is highly encouraging and supports the claim that most of the sequence-based splicing logic is located proximal to the splice sites. It is also noteworthy that TrASPr performs well for exons longer than 200, suggesting that most of the "regulatory code" is present at the exon boundaries rather than in its center (which TrASPr is blind to).

Additionally, Pearson correlation is used as the sole performance metric in many analyses (e.g., Fig 2 - Supp 2). The authors should consider alternative accuracy metrics, such as RMSE, which better convey the magnitude of prediction error and are more easily comparable across datasets. Pearson correlation may also be more sensitive to outliers on the smaller samples that arise when binning sequences.

- L171, "training it on cassette exons". This seems like an important point: previous models were trained mostly on constitutive exons, whereas here the model is trained specifically on cassette exons. This should be discussed in more detail.

* Our initial comment was incorrect, as pointed out by the authors. *

- L214, ablations of individual features are missing.

* This was addressed in the revision. *

- L230, "ENCODE cell lines", it is not clear why other tissues from GTEx were not included

* This was addressed in the revision. *

- L239, it is surprising that SpliceAI performs so badly, and might suggest a mistake in the analysis. Additional analysis and possible explanations should be provided to support these claims. Similarly for the complete failure of SpliceAI and Pangolin shown in Fig 4d.

* The authors should consider adding SpliceAI/Pangolin predictions for the alternative 5' and 3' splice site selection tasks (and code for related analyses) to the BitBucket repository.*

- BOS seems like a separate contribution that belongs in a separate publication. Instead, consider providing more details on TrASPr.

*Minor comment added after revision: regarding the author response that "A completely independent evaluation would have required a high-throughput experimental system to assess designs, which is beyond the scope of the current paper.":

It's not clear why BOS cannot be evaluated as a separate contribution by instead using different "teacher" models instead of TrASPr. Additionally, BOS lacks evaluation against existing methods for sequence optimization. *

- The authors should consider evaluating BOS using Pangolin or SpliceTransformer as the oracle, in order to measure the contribution to the sequence generation task provided by BOS vs TrASPr.

* See comment above *

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Author response:

The following is the authors' response to the original reviews

A point by point response included below. Before we turn to that we want to note one change that we decided to introduce, related to generalization on unseen tissues/cell types (Figure 3a in the original submission and related question by Reviewer #2 below). This analysis was based on adding a latent "RBP state" representation during learning of condition/tissue specific splicing. The "RBP state" per condition is captured by a dedicated encoder. Our original plan was to have a paper describing a new RBP-AE model we developed in parallel, which also served as the base to capture this "RBP State". However, we got delayed in getting this second paper finalized (it was led by other lab members, some of whom have already left the lab). This delay affected the TrASPr manuscript as TrASPr's code should be available and analysis reproducible upon publication. After much deliberation, we decided that in order to comply with reproducibility standards while not self scooping the RBP-AE paper, we eventually decided to take out the RBP-AE and replace it with a vanilla PCA based embedding for the "RBP-State". The PCA approach is simpler and reproducible, based on linear transformation of the RBPs expression vector into a lower dimension. The qualitative results included in Figure 3a still hold, and we also produced the new results suggested by Reviewer #2 in other GTEx tissues with this PCA based embedding (below).

We don't believe the switch to PCA based embedding should have any bearing on the current manuscript evaluation but wanted to take this opportunity to explain the reasoning behind this additional change.

Public Reviews:

Reviewer #1 (Public review):

Summary:

The authors propose a transformer-based model for the prediction of condition - or tissue-specific alternative splicing and demonstrate its utility in the design of RNAs with desired splicing outcomes, which is a novel application. The model is compared to relevant existing approaches (Pangolin and SpliceAI) and the authors clearly demonstrate its advantage. Overall, a compelling method that is well thought out and evaluated.

Strengths:

(1) The model is well thought out: rather than modeling a cassette exon using a single generic deep learning model as has been done e.g. in SpliceAI and related work, the authors propose a modular architecture that focuses on different regions around a potential exon skipping event, which enables the model to learn representations that are specific to those regions. Because each component in the model focuses on a fixed length short sequence segment, the model can learn position-specific features. Another difference compared to Pangolin and SpliceAI which are focused on modeling individual splice junctions is the focus on modeling a complete alternative splicing event.

(2) The model is evaluated in a rigorous way - it is compared to the most relevant state-of-the-art models, uses machine learning best practices, and an ablation study demonstrates the contribution of each component of the architecture.

(3) Experimental work supports the computational predictions.

(4) The authors use their model for sequence design to optimize splicing outcomes, which is a novel application.

We wholeheartedly thank Reviewer #1 for these positive comments regarding the modeling approach we took to this task and the evaluations we performed. We have put a lot of work and thought into this and it is gratifying to see the results of that work acknowledged like this.

Weaknesses:

No weaknesses were identified by this reviewer, but I have the following comments:

(1) I would be curious to see evidence that the model is learning position-specific representations.

This is an excellent suggestion to further assess what the model is learning. To get a better sense of the position-specific representation we performed the following analyses:

(1) Switching the transformers relative order: All transformers are pretrained on 3' and 5' splice site regions before fine-tuning for the PSI and dPSI prediction task. We hypothesized that if relative position is important, switching the order of the transformers would make a large difference on prediction accuracy. Indeed if we switch the 3' and 5' we see as expected a severe drop in performance, with Pearson correlation on test data dropping from 0.82 to 0.11. Next, we switched the two 5' and 3' transformers, observing a drop to 0.65 and 0.78 respectively. When focusing only on changing events the drop was from 0.66 to 0.54 (for 3' SS transformers), 0.48 (for 5' SS transformers), and 0.13 (when the 3' and 5' transformers flanking the alternative exon were switched).

(2) Position specific effect of RBPs: We wanted to test whether the model is able to learn position specific effects for RBPs. For this we focused on two RBPs, FOX (a family of three highly related RBPs), and QKI, both have a relatively well defined motif, known condition and position specific effect identified via RBP KD experiments combined with CLIP experiments (e.g. PMID: 23525800, PMID: 24637117, PMID: 32728246). For each, we randomly selected 40 highly and 40 lowly included cassette exons sequences. We then ran in-silico mutagenesis experiments where we replaced small windows of sequences with the RBP motifs (80 for RBOX and 80 for QKI), then compared TrASPR's predictions for the average predictions for 5 random sequences inserted in the same location. The results of this are now shown in Figure 4 Supp 3, where the y-axis represents the dPSI effect per position (x-axis), and the color represents the percentile of observed effects over inserting motifs in that position across all 80 sequences tested. We see that both RBPs have strong positional preferences for exerting a strong effect on the alternative exon. We also see differences between binding upstream and

downstream of the alternative exon. These results, learned by the model from natural tissue-specific variations, recapitulate nicely the results derived from high-throughput experimental assays. However, we also note that effects were highly sequence specific. For example, RBFOX is generally expected to increase inclusion when binding downstream of the alternative exon and decrease inclusion when binding upstream. While we do observe such a trend we also see cases where the opposite effects are observed. These sequence specific effects have been reported in the literature but may also represent cases where the model errs in the effect's direction. We discuss these new results in the revised text.

(3) Assessing BOS sequence edits to achieve tissue-specific splicing: Here we decided to test whether BOS edits in intronic regions (at least 8b away from the nearest splice site) are important for the tissue-specific effect. The results are now included in Figure 6 Supp 1, clearly demonstrating that most of the neuronal specific changes achieved by BOS were based on changing the introns, with a strong effect observed for both up and downstream intron edits.

(2) The transformer encoders in TrASPr model sequences with a rather limited sequence size of 200 bp; therefore, for long introns, the model will not have good coverage of the intronic sequence. This is not expected to be an issue for exons.

The reviewer is raising a good question here. On one hand, one may hypothesize that, as the reviewer seems to suggest, TrASPr may not do well on long introns as it lacks the full intronic sequence.

Conversely, one may also hypothesize that for long introns, where the flanking exons are outside the window of SpliceAI/Pangolin, TrASPr may have an advantage.

Given this good question and a related one by Reviewer #2, we divided prediction accuracy by intron length and the alternative exon length.

For short exons (<100bp) we find TrASPr and Pangolin perform similarly, but for longer exons, especially those > 200, TrASPr results are better. When dividing samples by the total length of the upstream and downstream intron, we find TrASPr outperform all other models for introns of combined length up to 6K, but Pangolin gets better results when the combined intron length is over 10K. This latter result is interesting as it means that contrary to the second hypothesis laid out above, Pangolin's performance did not degrade for events where the flanking exons were outside its field of view. We note that all of the above holds whether we assess all events or just cases of tissue specific changes. It is interesting to think about the mechanistic causes for this. For example, it is possible that cassette exons involving very long introns evoke a different splicing mechanism where the flanking exons are not as critical and/or there is more signal in the introns which is missed by TrASPr. We include these new results now as Figure 2 - Supp 1,2 and discuss these in the main text.

(3) In the context of sequence design, creating a desired tissue- or condition-specific effect would likely require disrupting or creating motifs for splicing regulatory proteins. In your experiments for neuronal-specific Daam1 exon 16, have you seen evidence for that? Most of the edits are close to splice junctions, but a few are further away.

That is another good question. Regarding Daam1 exon 16, in the original paper describing the mutation locations some motif similarities were noted to PTB (CU) and CUG/Mbnl-like elements (Barash et al Nature 2010). In order to explore this question beyond this specific case we assessed the importance of intronic edits by BOS to achieve a tissue specific splicing profile - see above.

(4) For sequence design, of tissue- or condition-specific effect in neuronal-specific Daam1 exon 16 the upstream exonic splice junction had the most sequence edits. Is that a

general observation? How about the relative importance of the four transformer regions in TrASPr prediction performance?

This is another excellent question. Please see new experiments described above for RBP positional effect and BOS edits in intronic regions which attempt to give at least partial answers to these questions. We believe a much more systematic analysis can be done to explore these questions but such evaluation is beyond the scope of this work.

(5) *The idea of lightweight transformer models is compelling, and is widely applicable. It has been used elsewhere. One paper that came to mind in the protein realm:*

Singh, Rohit, et al. "Learning the language of antibody hypervariability." Proceedings of the National Academy of Sciences 122.1 (2025): e2418918121.

We definitely do not make any claim this approach of using lighter, dedicated models instead of a large 'foundation' model has not been taken before. We believe Rohit et al mentioned above represents a somewhat different approach, where their model (AbMAP) fine-tunes large general protein foundational models (PLM) for antibody-sequence inputs by supervising on antibody structure and binding specificity examples. We added a description of this modeling approach citing the above work and another one which specifically handles RNA splicing (intron retention, PMID: 39792954).

Reviewer #2 (Public review):

Summary:

The authors present a transformer-based model, TrASPr, for the task of tissue-specific splicing prediction (with experiments primarily focused on the case of cassette exon inclusion) as well as an optimization framework (BOS) for the task of designing RNA sequences for desired splicing outcomes.

For the first task, the main methodological contribution is to train four transformer-based models on the 400bp regions surrounding each splice site, the rationale being that this is where most splicing regulatory information is. In contrast, previous work trained one model on a long genomic region. This new design should help the model capture more easily interactions between splice sites. It should also help in cases of very long introns, which are relatively common in the human genome.

TrASPr's performance is evaluated in comparison to previous models (SpliceAI, Pangolin, and SpliceTransformer) on numerous tasks including splicing predictions on GTEx tissues, ENCODE cell lines, RBP KD data, and mutagenesis data. The scope of these evaluations is ambitious; however, significant details on most of the analyses are missing, making it difficult to evaluate the strength of the evidence. Additionally, state-of-the-art models (SpliceAI and Pangolin) are reported to perform extremely poorly in some tasks, which is surprising in light of previous reports of their overall good prediction accuracy; the reasoning for this lack of performance compared to TrASPr is not explored.

In the second task, the authors combine Latent Space Bayesian Optimization (LSBO) with a Transformer-based variational autoencoder to optimize RNA sequences for a given splicing-related objective function. This method (BOS) appears to be a novel application of LSBO, with promising results on several computational evaluations and the potential to be impactful on sequence design for both splicing-related objectives and other tasks.

We thank Reviewer #2 for this detailed summary and positive view of our work. It seems the main issue raised in this summary regards the evaluations: The reviewer finds details of the evaluations missing and the fact that SpliceAI and Pangolin perform poorly on some of the tasks to be surprising. We made a concise effort to include the required details, including

code and data tables. In short, some of the concerns were addressed by adding additional evaluations, some by clarifying missing details, and some by better explaining where Pangolin and SpliceAI may excel vs. settings where these may not do as well. More details are given below.

Strengths:

(1) A novel machine learning model for an important problem in RNA biology with excellent prediction accuracy.

(2) Instead of being based on a generic design as in previous work, the proposed model incorporates biological domain knowledge (that regulatory information is concentrated around splice sites). This way of using inductive bias can be important to future work on other sequence-based prediction tasks.

Weaknesses:

(1) Most of the analyses presented in the manuscript are described in broad strokes and are often confusing. As a result, it is difficult to assess the significance of the contribution.

We made an effort to make the tasks be specific and detailed, including making the code and data of those available. We believe this helped improve clarity in the revised version.

(2) As more and more models are being proposed for splicing prediction (SpliceAI, Pangolin, SpliceTransformer, TrASPr), there is a need for establishing standard benchmarks, similar to those in computer vision (ImageNet). Without such benchmarks, it is exceedingly difficult to compare models. For instance, Pangolin was apparently trained on a different dataset (Cardoso-Moreira et al. 2019), and using a different processing pipeline (based on SpliSER) than the ones used in this submission. As a result, the inferior performance of Pangolin reported here could potentially be due to subtle distribution shifts. The authors should add a discussion of the differences in the training set, and whether they affect your comparisons (e.g., in Figure 2). They should also consider adding a table summarizing the various datasets used in their previous work for training and testing. Publishing their training and testing datasets in an easy-to-use format would be a fantastic contribution to the community, establishing a common benchmark to be used by others.

There are several good points to unpack here. Starting from the last one, we very much agree that a standard benchmark will be useful to include. For tissue specific splicing quantification we used the GTEx dataset from which we select six representative human tissues (heart, cerebellum, lung, liver, spleen, and EBV-transformed lymphocytes). In total, we collected 38394 cassette exon events quantified across 15 samples (here a ‘sample’ is a cassette exon quantified in two tissues) from the GTEx dataset with high-confidence quantification for their PSIs based on MAJIQ. A detailed description of how this data was derived is now included in the Methods section, and the data itself is made available via the bitbucket repository with the code.

Next, regarding the usage of different data and distribution shifts for Pangolin: The reviewer is right to note there are many differences between how Pangolin and TrASPr were trained. This makes it hard to determine whether the improvements we saw are not just a result of different training data/labels. To address this issue, we first tried to finetune the pre-trained Pangolin with MAJIQ’s PSI dataset: we use the subset of the GTEx dataset described above, focusing on the three tissues analyzed in Pangolin’s paper—heart, cerebellum, and liver—for a fair comparison. In total, we obtained 17,218 events, and we followed the same training and test split as reported in the Pangolin paper. We got Pearson: 0.78 Spearman: 0.68 which are values similar to what we got without this extra fine tuning. Next, we retrained Pangolin

from scratch, with the full tissues and training set used for TrASPr, which was derived from MAJIQ's quantifications. Since our model only trained on human data with 6 tissues at the same time, we modified Pangolin from original 4 splice site usage outputs to 6 PSI outputs. We tried to take the sequence centered with the first or the second splice site of the mid exon. This test resulted in low performance (3' SS: pearson 0.21 5' SS: 0.26.).

The above tests are obviously not exhaustive but their results suggest that the differences we observe are unlikely to be driven by distribution shifts. Notably, the original Pangolin was trained on much more data (four species, four tissues each, and sliding windows across the entire genome). This training seems to be important for performance while the fact we switched from Pangolin's splice site usage to MAJIQ's PSI was not a major contributor. Other potential reasons for the improvements we observed include the architecture, target function, and side information (see below) but a complete delineation of those is beyond the scope of this work.

(3) Related to the previous point, as discussed in the manuscript, SpliceAI, and Pangolin are not designed to predict PSI of cassette exons. Instead, they assign a "splice site probability" to each nucleotide. Converting this to a PSI prediction is not obvious, and the method chosen by the authors (averaging the two probabilities (?)) is likely not optimal. It would be interesting to see what happens if an MLP is used on top of the four predictions (or the outputs of the top layers) from SpliceAI/Pangolin. This could also indicate where the improvement in TrASPr comes from: is it because TrASPr combines information from all four splice sites? Also, consider fine-tuning Pangolin on cassette exons only (as you do for your model).

Please see the above response. We did not investigate more sophisticated models that adjust Pangolin's architecture further as such modifications constitute new models which are beyond the scope of this work.

(4) L141, "TrASPr can handle cassette exons spanning a wide range of window sizes from 181 to 329,227 bases - thanks to its multi-transformer architecture." This is reported to be one of the primary advantages compared to existing models. Additional analysis should be included on how TrASPr performs across varying exon and intron sizes, with comparison to SpliceAI, etc.

This was a good suggestion, related to another comment made by Reviewer #1. Please see above our response to them with a breakdown by exon/intron length.

(5) L171, "training it on cassette exons". This seems like an important point: previous models were trained mostly on constitutive exons, whereas here the model is trained specifically on cassette exons. This should be discussed in more detail.

Previous models were not trained exclusively on constitutive exons and Pangolin specifically was trained with their version of junction usage across tissues. That said, the reviewer's point is valid (and similar to ones made above) about a need to have a matched training/testing and potential distribution shifts. Please see response and evaluations described above.

(6) L214, ablations of individual features are missing.

These were now added to the table which we moved to the main text (see table also below).

(7) L230, "ENCODE cell lines", it is not clear why other tissues from GTEx were not included.

Good question. The task here was to assess predictions in unseen conditions, hence we opted to test on completely different data of human cell lines rather than additional tissue samples. Following the reviewers suggestion we also evaluated predictions on two additional GTEx

tissues, Cortex and Adrenal Gland. These new results, as well as the previous ones for ENCODE, were updated to use the PCA based embedding of “RBP-State” as described above. We also compared the predictions using the PCA based embedding of the “RBP-State” to training directly on data (not the test data of course) from these tissues. See updated Figure 3a,b. Figure 3 Supp 1,2.

(8) L239, it is surprising that SpliceAI performs so badly, and might suggest a mistake in the analysis. Additional analysis and possible explanations should be provided to support these claims. Similarly, the complete failure of SpliceAI and Pangolin is shown in Figure 4d.

Line 239 refers to predicting relative inclusion levels between competing 3' and 5' splice sites. We admit we too expected this to be better for SpliceAI and Pangolin but we were not able to find bugs in our analysis (which is all made available for readers and reviewers alike). Regarding this expectation to perform better, first we note that we are not aware of a similar assessment being done for either of those algorithms (i.e. relative inclusion for 3' and 5' alternative splice site events). Instead, our initial expectation, and likely the reviewer's as well, was based on their detection of splice site strengthening/weakening due to mutations, including cryptic splice site activation. More generally though, it is worth noting in this context that given how SpliceAI, Pangolin and other algorithms have been presented in papers/media/scientific discussions, we believe there is a potential misperception regarding tasks that SpliceAI and Pangolin excel at vs other tasks where they should not necessarily be expected to excel. Both algorithms focus on cryptic splice site creation/disruption. This has been the focus of those papers and subsequent applications. While Pangolin added tissue specificity to SpliceAI training, the authors themselves admit “...predicting differential splicing across tissues from sequence alone is possible but remains a considerable challenge and requires further investigation”. The actual performance on this task is not included in Pangolin's main text, but we refer Reviewer #2 to supplementary figure S4 in the Pangolin manuscript to get a sense of Pangolin's reported performance on this task. Similar to that, Figure 4d in our manuscript is for predicting ‘tissue specific’ regulators. We do not think it is surprising that SpliceAI (tissue agnostic) and Pangolin (slight improvement compared to SpliceAI in tissue specific predictions) do not perform well on this task. Similarly, we do not find the results in Figure 4C surprising either. These are for mutations that slightly alter inclusion level of an exon, not something SpliceAI was trained on - SpliceAI was trained on genomic splice sites with yes/no labels across the genome. As noted elsewhere in our response, re-training Pangolin on this mutagenesis dataset results in performance much closer to that of TrASPr. That is to be expected as well - Pangolin is constructed to capture changes in PSI (or splice site usage as defined by the authors), those changes are not even tissue specific for the CD19 data and the model has no problem/lack of capacity to generalize from the training set just like TrASPr does. In fact, if you only use combinations of known mutations seen during training a simple regression model gives correlation of ~92-95% (Cortés-López et al 2022). In summary, we believe that better understanding of what one can realistically expect from models such as SpliceAI, Pangolin, and TrASPr will go a long way to have them better understood and used effectively. We have tried to make this more clear in the revision.

(9) BOS seems like a separate contribution that belongs in a separate publication. Instead, consider providing more details on TrASPr.

We thank the reviewer for the suggestion. We agree those are two distinct contributions/algorithms and we indeed considered having them as two separate papers. However, there is strong coupling between the design algorithm (BOS) and the predictor that enables it (TrASPr). This coupling is both conceptual (TrASPr as a “teacher”) and practical in terms of evaluations. While we use experimental data (experiments done involving Daam1 exon 16, CD19 exon 2) we still rely heavily on evaluations by TrASPr itself. A completely

independent evaluation would have required a high-throughput experimental system to assess designs, which is beyond the scope of the current paper. For those reasons we eventually decided to make it into what we hope is a more compelling combined story about generative models for prediction and design of RNA splicing.

(10) The authors should consider evaluating BOS using Pangolin or SpliceTransformer as the oracle, in order to measure the contribution to the sequence generation task provided by BOS vs TrASPr.

We can definitely see the logic behind trying BOS with different predictors. That said, as we note above most of BOS evaluations are based on the “teacher”. As such, it is unclear what value replacing the teacher would bring. We also note that given this limitation we focus mostly on evaluations in comparison to existing approaches (genetic algorithm or random mutations as a strawman).

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Additional comments:

(1) Is your model picking up transcription factor binding sites in addition to RBPs? TFs have been recently shown to have a role in splicing regulation:

Daoud, Ahmed, and Asa Ben-Hur. "The role of chromatin state in intron retention: A case study in leveraging large scale deep learning models." PLOS Computational Biology 21.1 (2025): e1012755.

We agree this is an interesting point to explore, especially given the series of works from the Ben-Hur’s group. We note though that these works focus on intron retention (IR) which we haven’t focused on here, and we only cover short intronic regions flanking the exons. We leave this as a future direction as we believe the scope of this paper is already quite extensive.

(2) SpliceNouveau is a recently published algorithm for the splicing design problem:

Wilkins, Oscar G., et al. "Creation of de novo cryptic splicing for ALS and FTD precision medicine." Science 386.6717 (2024): 61-69.

Thank you for pointing out Wilkins et al recent publication, we now refer to it as well.

(3) Please discuss the relationship between your model and this deep learning model. You will also need to change the following sentence: "Since the splicing sequence design task is novel, there are no prior implementations to reference."

We revised this statement and now refer to several recent publications that propose similar design tasks.

(4) I would suggest adding a histogram of PSI values - they appear to be mostly close to 1 or 0.

PSI values are indeed typically close to either 0 or 1. This is a known phenomenon illustrated in previous studies of splicing (e.g. Shen et al NAR 2012). We are not sure what is meant by the comment to add a histogram but we made sure to point this out in the main text:

“...Still, those statistics are dominated by extreme values, such that 33.2\% are smaller than 0.15 and 56.0\% are higher than 0.85. Furthermore, most cassette exons do not change between a given tissue pair (only 14.0\% of the samples in the dataset, i.e a cassette exon measured across two tissues, exhibit $\Delta\Psi \geq 0.15$).”

(5) Part of the improvement of TrASPr over Pangolin could be the result of a more extensive dataset.

Please see above responses and new analysis.

(6) In the discussion of the roles of alternative splicing, protein diversity is mentioned, but I suggest you also mention the importance of alternative splicing as a regulatory mechanism:

Lewis, Benjamin P., Richard E. Green, and Steven E. Brenner. "Evidence for the widespread coupling of alternative splicing and nonsense-mediated mRNA decay in humans." *Proceedings of the National Academy of Sciences* 100.1 (2003): 189-192.

Thank you for the suggestion. We added that point and citation.

(7) Line 96: You use dPSI without defining it (although quite clear that it should be Delta PSI).

Fixed.

(8) Pretrained transformers: Have you trained separate transformers on acceptor and donor sites, or a single splice junction transformer?

Single splice junction pre-training.

(9) "TrASPr measures the probability that the splice site in the center of S_e is included in some tissue" - that's not my understanding of what TrASPr is designed to do.

We revised the above sentence to make it more precise: "Given a genomic sequence context $S_e = (s_e, \dots, s_e)$, made of a cassette exon e and flanking intronic/exonic regions, TrASPr predicts for tissue c the fraction of transcripts where exon e is included or skipped over, $\Delta\Psi_{-e,c,c'}$."

(10) Please include the version of the human genome annotations that you used.

We used GENCODE v40 human genome hg38- this is now included in the Data section.

(11) I did not see a description of the RBP-AE component in the methods section. A bit more detail on the model would be useful as well.

Please see above details about replacing RBP-AE with a simpler linear PCA "RBP-State" encoding. We added details about how the PCA was performed to the Methods section.

(12) Typos, grammar:

- Fix the following sentence: ATP13A2, a lysosomal transmembrane cation transporter, linked to an early-onset form of Parkinson's Disease (PD) when 306 loss-of-function mutations disrupt its function.

Sentence was fixed to now read: "The first example is of a brain cerebellum-specific cassette exon skipping event predicted by TrASPr in the ATP13A2 gene (aka PARK9). ATP13A2 is a lysosomal transmembrane cation transporter, for which loss of function mutation has been linked to early-onset of Parkinson's Disease (PD)".

- Line 501: "was set to $4e-4$ " (the e is a superscript).

Fixed

- A couple of citations are missing in lines 580 and 581.

Thank you for catching this error. Citations in line 580, 581 were fixed.

(13) Paper title: *Generative modeling for RNA splicing predictions and design - it would read better as "Generative modeling for RNA splicing prediction and design", as you are solving the problems of splicing prediction and splicing design.*

Thank you for the suggestion. We updated the title and removed the plural form.

Reviewer #2 (Recommendations for the authors):

(1) Appendices are not very common in biology journals. It is also not clear what purpose the appendix serves exactly - it seems to repeat some of the things said earlier. Consider merging it into the methods or the main text.

We merged the appendices into the Methods section and removed redundancy.

(2) L112, "For instance, the model could be tasked with designing a new version of the cassette exon, restricted to no more than N edit locations and M total base changes." How are N and M different? Is there a difference between an edit location and a base change?

Yes, N is the number of locations (one can think of it as a start position) of various lengths (e.g. a SNP is of length 1) and the total number of positions edited is M . The text now reads "For instance, the model could be tasked with designing a new version of the cassette exon, restricted to no more than N edit locations (ie start position of one or more consecutive bases) and M total base changes."

(3) L122: "DEN was developed for a distinct problem". What prevents one from adapting DEN to your sequence design task? The method should be generic. I do not see what "differs substantially" means here. (Finally, wasn't DEN developed for the task you later refer to as "alternative splice site" (as opposed to "splice site selection")? Use consistent terminology. And in L236 you use "splice site variation" - is that also the same?).

Indeed, our original description was not clear/precise enough. DEN was designed and trained for two tasks: APA, and 5' alternative splice site usage. The terms "selection", "usage", and "variation" were indeed used interchangeably in different locations and the reviewer was right, noting the lack of precision. We have now revised the text to make sure the term "relative usage" is used.

Nonetheless, we hold DEN was indeed defined for different tasks. See figures from Figure 2A, 6A of Linder et al 2020 (the reference was also incorrect as we cited the preprint and not the final paper):

In both cases DEN is trying to optimize a short region for selecting an alternative PA site (left) or a 5' splice site (right). This work focused on an MPRA dataset of short synthetic sequences inserted in the designated region for train/test. We hold this is indeed a different type of data and task then the one we focus on here. Yes, one can potentially adopt DEN for our task, but this is beyond the scope of this paper. Finally, we note that a more closely related algorithm recently proposed is Ledidi (Schreiber et al 2025) which was posted as a pre-print. Similar to BOS, Ledidi tries to optimize a given sequence and adopt it with a few edits for a given task. Regardless, we updated the main text to make the differences between DEN and the task we defined here for BOS more clear, and we also added a reference to Ledidi and other recent works in the discussion section.

(4) L203, exons with DeltaPSI very close to 0.15 are going to be nearly impossible to classify (or even impossible, considering that the DeltaPSI measurements are not perfect). Consider removing such exons to make the task more feasible.

Yes, this is how it was done. As described in more details below, we defined changing samples as ones where the change was ≥ 0.15 and non-changing as ones where the change in PSI was < 0.05 to avoid ambiguous cases affecting the classification task.

(5) L230, *RBP-AE is not explained in sufficient detail (and does not appear in the methods, apparently). It is not clear how exactly it is trained on each new cellular condition.*

Please see response in the opening of this document and Q11 from

Reviewer 1

(6) L230, *"significantly improving": the r value actually got worse; it is therefore not clear you can claim any significant improvement. Please mention that fact in the text.*

This is a fair point. We note that we view the “a” statistic as potentially more interesting/relevant here as the Pearson “r” is dominated by points being generally close to 0/1. Regardless, revisiting this we realized one can also make a point that the term “significant” is imprecise/misplaced since there is no statistical test done here (side note: given the amount of points, a simple null of same distribution yes/no would pass significance but we don’t think this is an interesting/relevant test here). Also, we note that with the transition to PCA instead of RBP-AE we actually get improvements in both a and r values, both for the ENCODE samples shown in Figure 3a and the two new GTEx tissues we tested (see above). We now changed the text to simply state:

“...As shown in Figure 3a, this latent space representation allows TrSAPr to generalize from the six GTEx tissues to unseen conditions, including unseen GTEx tissues (top row), and ENCODE cell lines (bottom row). It improves prediction accuracy compared to TrASPr lacking PCA (eg a=88.5% vs a=82.3% for ENCODE cell lines), though naturally training on the additional GTEx and ENCODE conditions can lead to better performance (eg a=91.7%, for ENCODE, Figure 3a left column).”

(7) L233, *"Notably, previous splicing codes focused solely on cassette exons", Rosenberg et al. focused solely on alternative splice site choice.*

Right - we removed that sentence..

(8) L236, *"trained TrASPr on datasets for 3' and 5' splice site variations". Please provide more details on this task. What is the input to TrASPr and what is the prediction target (splice site usage, PSI of alternative isoforms)? What datasets are used for this task?*

The data for this data was the same GTEx tissue data processed, just for alternative 3’ and 5’ splice sites events. We revised the description of this task in the main task and added information in the Methods section. The data is also included in the repo.

(9) L243, *"directly from genomic sequences", and conservation?*

Yes, we changed the sentence to read “...directly from genomic sequences combined with related features”

(10) L262, *what is the threshold for significant splicing changes?*

The threshold is 0.15 We updated the main text to read the following:

The total number of mutations hitting each of the 1198 genomic positions across the 6106 sequences is shown in \FIG{mut_effect}b (left), while the distribution of effects (Δ | Ψ) observed across those 6106 samples is shown in \FIG{mut_effect}b (right). To this data we applied three testing schemes. The first is a standard 5-fold CV where 20% of combinations of point mutations were hidden in every fold while the second test involved

'unseen mutation' (UM) where we hide any sample that includes mutations in specific positions for a total of 1480 test samples. As illustrated by the CDF in FIG{mut_effect}b, most samples (each sample may involve multiple positions mutated) do not involve significant splicing changes. Thus, we also performed a third test using only the 883 samples where mutations cause significant changes ($|\Delta \Psi| \geq 0.15$).

(11) L266, Pangolin performance is only provided for one of the settings (and it is not clear which). Please provide details of its performance in all settings.

The description was indeed not clear. Pangolin's performance was similar to SpliceAI as mentioned above but retraining it on the CD19 data yielded much closer performance to TrASPr. We include all the matching tests for Pangolin after retraining in Figure 4 Supp Figure 1.

(12) Please specify "n=" in all relevant plots.

Fixed.

(13) Figure 3a, "The tissues were first represented as tokens, and new cell line results were predicted based on the average over conditions during training." Please explain this procedure in more detail. What are these tokens and how are they provided to the model? Are the cell line predictions the average of the predictions for the training tissues?

Yes, we compared to simply the average over the predictions for the training tissues for that specific event as baseline to assess improvements (see related work pointing for the need to have similar baselines in DL for genomics in <https://pubmed.ncbi.nlm.nih.gov/33213499/>). Regarding the tokens - we encode each tissue type as a possible value and feed the two tissues as two tokens to the transformer.

(14) Figure 4b, the total count in the histogram is much greater than 6106. Please explain the dataset you're using in more detail, and what exactly is shown here.

We updated the text to read:

"...we used 6106 sequence samples where each sample may have multiple positions mutated (ie mutation combinations) in exon 2 of CD19 and its flanking introns and exons (Cortes et al 2022). The total number of mutations hitting each of the 1198 genomic positions across the 6106 sequences is shown in Figure 4b (left)."

(15) Figure 5a, how are the prediction thresholds (TrASPr passed, TrASPr stringent, and TrASPr very stringent) defined?

Passed: $dpsi > 0.1$, Stringent: $dpsi > 0.15$, Very stringent: $dpsi > 0.2$ This is now included in the main text.

(16) L417, please include more detail on the relative size of TrASPr compared to other models (e.g. number of parameters, required compute, etc.).

SpliceAI is a general-purpose splicing predictor with 32-layer deep residual neural network to capture long-range dependencies in genomic sequences. Pangolin is a deep learning model specifically designed for predicting tissue-specific splicing with similar architecture as SpliceAI. The implementation of SpliceAI that can be found here <https://huggingface.co/multimolecule/spliceai> involves an ensemble of 5 such models for a total of ~3.5M parameters. TrASPr, has 4 BERT transformers (each 6 layers and 12 heads) and MLP a top of those for a total of ~189M parameters. Evo 2, a genomic 'foundation' model has 40B parameters, DNABERT has ~86M (a single BERT with 12 layers and 12 heads), and Borzoi has 186M parameters (as stated in

<https://www.biorxiv.org/content/10.1101/2025.05.26.656171v2> [↗](#)). We note that the difference here is not just in model size but also the amount of data used to train the model. We edited the original L417 to reflect that.

(17) L546, please provide more detail on the VAE. What is the dimension of the latent representation?

We added more details in the Methods section like the missing dimension (256) and definitions for P(Z) and P(S).

(18) Consider citing (and possibly comparing BOS to) Ghari et al., NeurIPS 2024 ("GFlowNet Assisted Biological Sequence Editing").

Added.

(19) Appendix Figure 2, and corresponding main text: it is not clear what is shown here. What is dPSI+ and dPSI-? What pairs of tissues are you comparing? Spearman correlation is reported instead of Pearson, which is the primary metric used throughout the text.

The dPSI+ and dPSI- sets were indeed not well defined in the original submission. Moreover, we found our own code lacked consistency due to different tests executed at different times/by different people. We apologize for this lack of consistency and clarity which we worked to remedy in the revised version. To answer the reviewer's question, given two tissues (c_1, c_2), dPSI+ and dPSI- is for correctly classifying the exons that are significantly differentially included or excluded. Specifically, differential included exons are those for which $|\Delta \Psi_{e,c_1,c_2}| = |\Psi_{e,c_1} - \Psi_{e,c_2}| \geq 0.15$, compared to those that are not ($|\Delta \Psi_{e,c_1,c_2}| < 0.05$). Similarly, dPSI- is for correctly classifying the exons that are significantly differentially excluded in the first tissue or included in the second tissue ($|\Delta \Psi_{e,c_1,c_2}| = |\Psi_{e,c_1} - \Psi_{e,c_2}| \leq -0.15$) compared to those that are not ($|\Delta \Psi_{e,c_1,c_2}| > -0.05$). This means dPSI+ and dPSI- are dependent on the order of c_1, c_2 . In addition, we also define a direction/order agnostic test for changing vs non changing events i.e. $|\Delta \Psi_{e,c_1,c_2}| \geq 0.15$ vs $|\Delta \Psi_{e,c_1,c_2}| < 0.05$. These test definitions are consistent with previous publications (e.g. Barash et al Nature 2010, Jha et al 2017) and also answer different biological questions: For example "Exons that go up in brain" and "Exons that go up in Liver" can reflect distinct mechanisms, while changing exons capture a model's ability to identify regulated exons even if the direction of prediction may be wrong. The updated Appendix Figure 2 is now in the main text as Figure 2d and uses Pearson, while AUPRC and AUROC refer to the changing vs no-changing classification task described above such that we avoid dPSI+ and dPSI- when summarizing in this table over 3 pairs of tissues. Finally, we note that making sure all tests comply with the above definition also resulted in an update to Figure 2b/c labels and values, where TrASPr's improvements over Pangolin reaches up to 1.8fold in AUPRC compared to 2.4fold in the earlier version. We again apologize for having a lack of clarity and consistent evaluations in the original submission.

(20) Minor typographical comments:

- Some plots could use more polishing (e.g., thicker stroke, bigger font size, consistent style (compare 4a to the other plots)...).

Agreed. While not critical for the science itself we worked to improve figure polishing in the revision to make those more readable and pleasant.

- Consider using 2-dimensional histograms instead of the current kernel density plots, which tend to over-smooth the data and hide potentially important details.

We were not sure what the exact suggestion is here and opted to leave the plots as is.

| - L53: $dPSI_{\{e, c, c'\}}$ is never formally defined. Is it $PSI_{\{e, c\}} - PSI_{\{e, c'\}}$ or vice versa?

Definition now included (see above).

| - L91: Define/explain "transformer" and provide reference.

We added the explanation and related reference of the transformer in the introduction section and BERT in the method section.

| - L94: exons are short. Are you referring here to the flanking introns? Please explain.

We apologize for the lack of clarity. We are referring to a cassette exon alternative splicing event as is commonly defined by the splice junctions involved that is from the 5' SS of the upstream exon to the 3' SS of the downstream exon. The text now reads:

"...In contrast, 24% of the cassette exons analyzed in this study span a region between the flanking exons' upstream 3' and downstream 5' splice sites that are larger than 10 kb."

| - L132: It's unclear whether a single, shared transformer or four different transformers (one for each splice site) are being pre-trained. One would at least expect 5' and 3' splice sites to have a different transformer. In Methods, L506, it seems that each transformer is pre-trained separately.

We updated the text to read:

"We then center a dedicated transformer around each of the splice sites of the cassette exon and its upstream and downstream (competing) exons (four separate transformers for four splice sites in total)."

| - L471: You explain here that it is unclear what tasks 'foundation' models are good for. Also in L128, you explain that you are not using a 'foundation' model. But then in L492, you describe the BERT model you're using as a foundation model!

Line 492 was simply a poor choice of wording as "foundation" is meant here simply as the "base component". We changed it accordingly.

| - L169, "pre-training ... BERT", explain what exactly this means. Is it using masking? Is it self-supervised learning? How many splice sites do you provide? Also explain more about the BERT architecture and provide references.

We added more details about the BERT architecture and training in the Methods section.

| - L186 and later, the values for a and r provided here and in the below do not correspond to what is shown in Figure 2.

Fixed, thank you for noticing this.

| - L187,188: What exactly do you mean by "events" and "samples"? Are they the same thing? If so, are they (exon, tissue) pairs? Please use consistent terminology. Moreover, when you say "changing between two conditions": do you take all six tissues whenever there is a 0.15 spread in PSI among them? Or do you take just the smallest PSI tissue and the largest PSI tissue when there is a 0.15 spread between them? Or something else altogether?

Reviewer #2 is yet again correct that the definitions were not precise. A "sample" involves a specific exon skipping "event" measured in two tissues. The text now reads:

"...most cassette exons do not change between a given tissue pair (only 14.0% of the samples in the dataset, i.e., a cassette exon measured across two tissues, exhibit $|\Delta\Psi| \geq 0.15$). Thus, when we repeat this analysis only for samples involving exons that exhibited a change in inclusion ($|\Delta\Psi| \geq 0.15$) between at least two tissues, performance degrades for all three models, but the differences between them become more striking (Figure 2a, right column)."

- Figure 1a, explain the colors in the figure legend. The 3D effect is not needed and is confusing (ditto in panel C).

Color explanation is now added: "exons and introns are shown as blue rectangles and black lines. The blue dashed line indicates the inclusive pattern and the red junction indicates an alternative splicing pattern."

These are not 3D effects but stacks to indicate multiple events/cases. We agree these are not needed in Fig1a to illustrate types of AS and removed those. However, in Fig1c and matching caption we use the stacks to indicate HT data captures many such LSVs over which ML algorithms can be trained.

- Figure 1b, this cartoon seems unnecessary and gives the wrong impression that this paper explores mechanistic aspects of splicing. The only relevant fact (RBPs serving as splicing factors) can be explained in the text (and is anyway not really shown in this figure).

We removed Figure 1b cartoon.

- Figure 1c, what is being shown by the exon label "8"?

This was meant to convey exon ID, now removed to simplify the figure.

- Figure 1e, left, write "Intron Len" in one line. What features are included under "..."? Based on the text, I did not expect more features.

Also, the arrows emanating from the features do not make sense. Is "Embedding" a layer? I don't think so. Do not show it as a thin stripe. Finally, what are dPSI'+ and dPSI'-? are those separate outputs? are those logits of a classification task?

We agree this description was not good and have updated it in the revised version.

- Figure 1e, the right-hand side should go to a separate figure much later, when you introduce BOS.

We appreciate the suggestion. However, we feel that Figure 1e serves as a visual representation of the entire framework. Just like we opted to not turn this work into two separate papers (though we fully agree it is a valid option that would also increase our publication count), we also prefer to leave this unified visual representation as is.

- Figure 2, does the $n=2456$ refer to the number of (exons, tissues) pairs? So each exon contributes potentially six times to this plot? Typo "approximately".

The "n" refers to the number of samples which is a cassette event measured in two tissues. The same cassette event may appear in multiple samples if it was confidently quantified in more than two tissues. We updated the caption to reflect this and corrected the typo.

- Figure 2b, typo "differentially included (dPSI+) or excluded" .

Fixed.

| - L221, "the DNABERT" => "DNABERT".

Fixed.

| - L232, missing percent sign.

-

Fixed.

| - L246, "see Appendix Section 2 for details" seems to instead refer to the third section of the appendix.

We do not have this as an Appendix, the reference has been updated.

| - Figure 3, bottom panels, PSI should be "splice site usage"?

PSI is correct here - we hope the revised text/definitions make it more clear now.

| - Figure 3b: typo: "when applied to alternative alternative 3".

Fixed.

| - p252, "polypyrimidine" (no capitalization).

Fixed.

| - Strange capitalization of tissue names (e.g., "Brain-Cerebellum"). The tissue is called "cerebellum" without capitalization.

We used EBV (capital) for the abbreviation and lower case for the rest.

| - Figure 4c: "predicted usage" on the left but "predicted PSI" on the right.

Right. We opted to leave it as is since Pangolin and SpliceAI do predict their definition of "usage" and not directly PSI, we just measure correlations to observed PSI as many works have done in the past.

| - Figure 4 legend typo: "two three".

Fixed.

| - L351, typo: "an (unsupervised)" (and no need to capitalize Transformer).

Fixed.

| - L384, "compared to other tissues at least" => "compared to other tissues of at least".

Fixed.

| - L549, P(Z) and P(S) are not defined in the text.

Fixed.

| - L572, remove "Subsequently". Add missing citations at the end of the paragraph.

Fixed.

- L580-581, citations missing.

Fixed.

| - L584-585, typo: "high confidince predictions"

Fixed.

| - L659-660, BW-M and B-WM are both used. Typo?

Fixed.

- L895, "calculating the average of these two", not clear; please rewrite.

Fixed.

| - L897, "Transformer" and "BERT", do these refer to the same thing? Be consistent.

BOS is a transformer and not a BERT but TrASPr uses the BERT architecture. BERT is a type of transformer as the reviewer is surely well aware so the sentence is correct. Still, to follow the reviewer's recommendation for consistency/clarity we changed it here to state BERT.

| - Appendix Figure 5: The term *dPSI* appears to be overloaded to also represent the difference between predicted PSI and measured PSI, which is inconsistent with previous definitions.

Indeed! We thank the reviewer again for their sharp eye and attention to details that we missed. We changed Supp Figure 5, now Figure 4 Supplementary Figure 2, to $|\text{PSI}' - \text{PSI}|$ and defined those as the difference between TrASPr's predictions (PSI') and MAJIQ based PSI quantifications.

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